

COMPUTER NETWORKS – III Year B.Tech I Semester (R22)

Syllabus

UNIT - I

Network hardware, Network software, OSI, TCP/IP Reference models, Example Networks: ARPANET, Internet.

Physical Layer: Guided Transmission media: twisted pairs, coaxial cable, fiber optics, Wireless transmission. Data link layer: Design issues, framing, Error detection and correction.

UNIT - II

Elementary data link protocols: simplex protocol, A simplex stop and wait protocol for an error-free channel, A simplex stop and wait protocol for noisy channel.

Sliding Window protocols: A one-bit sliding window protocol, A protocol using Go-Back-N, A protocol using Selective Repeat, Example data link protocols.

Medium Access sub layer: The channel allocation problem, Multiple access protocols: ALOHA, Carrier sense multiple access protocols, collision free protocols. Wireless LANs, Data link layer switching.

UNIT - III

Network Layer: Design issues, Routing algorithms: shortest path routing, Flooding, Hierarchical routing, Broadcast, Multicast, distance vector routing, Congestion Control Algorithms, Quality of Service, Internetworking, The Network layer in the internet.

UNIT - IV

Transport Layer: Transport Services, Elements of Transport protocols, Connection management, TCP and UDP protocols.

UNIT - V

Application Layer –Domain name system, SNMP, Electronic Mail; the World WEB, HTTP, Streaming audio and video.

TEXT BOOK:

1. Computer Networks -- Andrew S Tanenbaum, David. j. Wetherall, 5th Edition. Pearson Education/PHI

REFERENCE BOOKS:

1. An Engineering Approach to Computer Networks-S. Keshav, 2nd Edition, Pearson Education
2. Data Communications and Networking – Behrouz A. Forouzan. Third Edition TMH.

Course Outcomes

Upon completion of this course, the student will be able to

C01: Classify different types of computer networks and illustrate the functions of network hardware, network software, OSI and TCP/IP reference models.

C02: Analyse flow control and error control mechanisms and apply them using standard data link layer protocols.

C03: Design subnets and apply various routing algorithms to find shortest paths for packet delivery

C04: Identify various Congestion control algorithms and calculate the IP addresses of the network

C05: Demonstrate the working of transport layer protocols, elements of transport protocols, connection management, and TCP & UDP services.

C06: Explain in detail the features and operations of various protocols of application layer such as HTTP, DNS and SMTP

Unit- I

Introduction

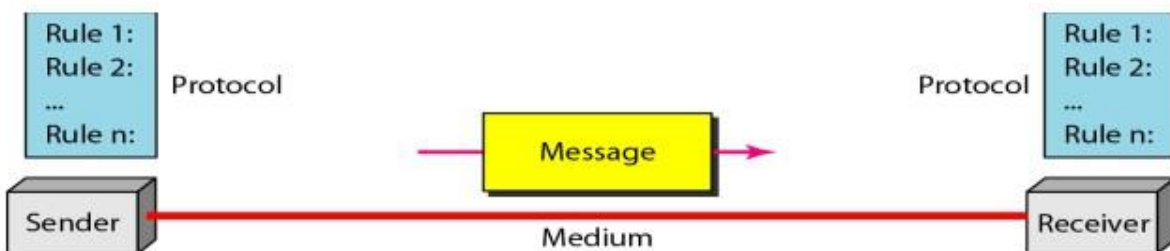
When we communicate, we are sharing information. This sharing can be local or remote. Between individuals, local communication usually occurs face to face, while remote communication takes place over distance.

The term *telecommunication*, which includes telephony, telegraphy, and television, means communication at a distance (*tele* is Greek for "far").

The word *data* refers to information presented in whatever form is agreed upon by the parties creating and using the data.

Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable. For data communications to occur, the communicating devices must be part of a communication system made up of a combination of hardware (physical equipment) and software (programs). The effectiveness of a data communications system depends on four fundamental characteristics: delivery, accuracy, timeliness, and jitter.

A data communications system has five components.



1. Message. The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.
2. Sender. The sender is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.
3. Receiver. The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.
4. Transmission medium. The transmission medium is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.

5. Protocol. A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.

Data Representation:

Information today comes in different forms such as text, numbers, images, audio, and

Video. Text

In data communications, text is represented as a bit pattern, a sequence of bits (0s or 1s). Different sets of bit patterns have been designed to represent text symbols. Each set is called a code, and the process of representing symbols is called coding. Today, the prevalent coding system is called Unicode, which uses 32 bits to represent a symbol or character used in any language in the world. The American Standard Code for Information Interchange (ASCII), developed some decades ago in the United States, now constitutes the first 127 characters in Unicode and is also referred to as Basic Latin.

Numbers

Numbers are also represented by bit patterns. However, a code such as ASCII is not used to represent numbers; the number is directly converted to a binary number to simplify mathematical operations. Appendix B discusses several different numbering systems.

Images

Images are also represented by bit patterns. In its simplest form, an image is composed of a matrix of pixels (picture elements), where each pixel is a small dot. The size of the pixel depends on the *resolution*. For example, an image can be divided into 1000 pixels or 10,000 pixels. In the second case, there is a better representation of the image (better resolution), but more memory is needed to store the image.

After an image is divided into pixels, each pixel is assigned a bit pattern. The size and the value of the pattern depend on the image. For an image made of only black and- white dots (e.g., a chessboard), a 1-bit pattern is enough to represent a pixel. If an image is not made of pure white and pure black pixels, you can increase the size of the bit pattern to include gray scale. For example, to show four levels of gray scale, you can use 2-bit patterns. A black pixel can be represented by 00, a dark gray pixel by 01, a light gray pixel by 10, and a white pixel by 11.

There are several methods to represent color images. One method is called RGB, so called because each color is made of a combination of three primary colors: *red*, green, and blue. The intensity of each color is measured, and a bit pattern is assigned to it. Another method is called YCM, in which a color is made of a combination of three other primary colors: yellow, cyan, and magenta.

Audio

Audio refers to the recording or broadcasting of sound or music. Audio is by nature different from text, numbers, or images. It is continuous, not discrete. Even when we use a microphone to change voice or music to an electric signal, we create a continuous signal.

Video

Video refers to the recording or broadcasting of a picture or movie. Video can either be produced as a continuous entity (e.g., by a TV camera), or it can be a combination of images, each a discrete entity, arranged to convey the idea of motion.

Data Flow:

Data flow (simplex, half-duplex, and full-duplex)



Simplex

In simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; the other can only receive. Keyboards and traditional monitors are examples of simplex devices. The keyboard can only introduce input; the monitor can only accept output. The simplex mode can use the entire capacity of the channel to send data in one direction.

Half-Duplex

In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa. The half-duplex mode is like a one-lane road with traffic allowed in both directions. When cars are traveling in one direction, cars going the other way must wait.

In a half-duplex transmission, the entire capacity of a channel is taken over by whichever of the two devices is transmitting at the time.

Example: Walkie-talkies and CB (citizens band) radios are both half-duplex systems. The half-duplex mode is used in cases where there is no need for communication in both directions at the same time; the entire capacity of the channel can be utilized for each direction.

Full-Duplex

In full-duplex mode, both stations can transmit and receive simultaneously. The full-duplex mode is like a two way street with traffic flowing in both directions at the same time. In full-duplex mode, signals going in one direction share the capacity of the link: with signals going in the other direction. This sharing can occur in two ways: Either the link must contain two physically separate transmission paths, one for sending and the other for receiving; or the capacity of the channel is divided between signals traveling in both directions.

One common example of full-duplex communication is the telephone network. When two people are communicating by a telephone line, both can talk and listen at the same time.

Network:

- A network is a set of devices (often referred to as *nodes*) connected by communication links. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.
- Most networks use distributed processing, in which a task is divided among multiple computers. Instead of one single large machine being responsible for all aspects of a process, separate computers (usually a personal computer or workstation) handle a subset.
- A network must be able to meet a certain number of criteria. The most important of these are performance, reliability, and security.

Performance

Performance can be measured in many ways, including transit time and response time. Transit time is the amount of time required for a message to travel from one device to another. Response time is the elapsed time between an inquiry and a response. The performance of a network depends on a number of factors, including the number of users, the type of transmission medium, the capabilities of the connected hardware, and the efficiency of the software.

Reliability

In addition to accuracy of delivery, network reliability is measured by the frequency of failure, the time it takes a link to recover from a failure, and the network's robustness in a catastrophe.

Security

Network security issues include protecting data from unauthorized access, protecting data from damage and development, and implementing policies and procedures for recovery from breaches and data losses.

Computer Networks:

A Computer Network is interconnection of several computers by communication links referred to as computer networks.

Advantages:

Resource Sharing

Ease of Accessibility

Less cost

Flexibility

Security

Type of Connection:

A network is two or more devices connected through links. A link is a communication pathway that transfers data from one device to another. For visualization purposes, it is simplest to imagine any link as a line drawn between two points. For communication to occur, two devices must be connected in some way to the same link at the same time. There are two possible types of connections:

1. Point-to-point connection
2. Multipoint connection

Point-to-Point

A point-to-point connection provides a dedicated link between two devices. The entire capacity of the link is reserved for transmission between those two devices.

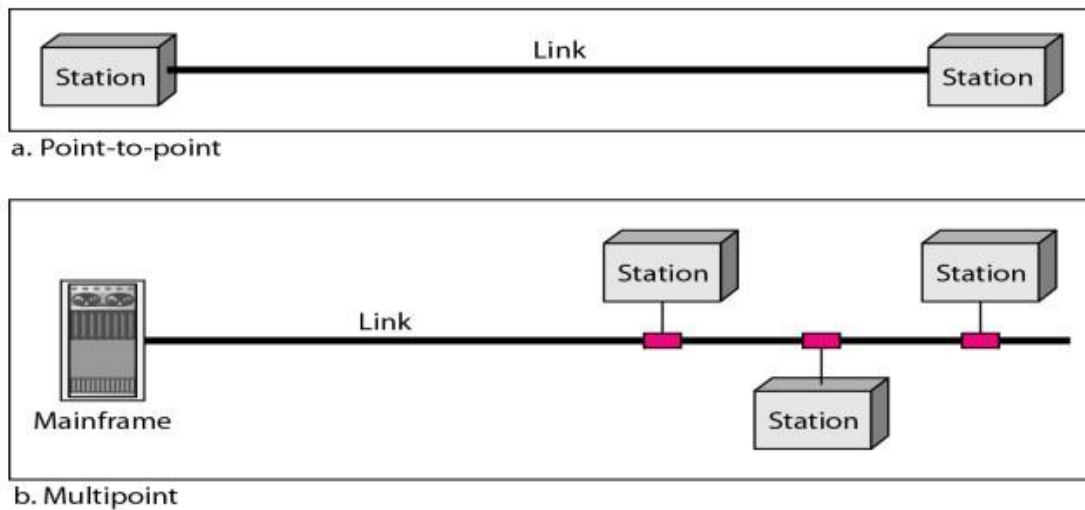
Most point-to-point connections use an actual length of wire or cable to connect the two ends, but other options, such as microwave or satellite links, are also possible.

When you change television channels by infrared remote control, you are establishing a point-to-point connection between the remote control and the television's control system.

Multipoint

A multipoint (also called multi drop) connection is one in which more than two specific devices share a single link. In a multipoint environment, the capacity of the channel is shared, either spatially or temporally. If several devices can use the link simultaneously, it is a spatially shared connection.

If users must take turns, it is a timeshared connection.



Computer Network Architecture

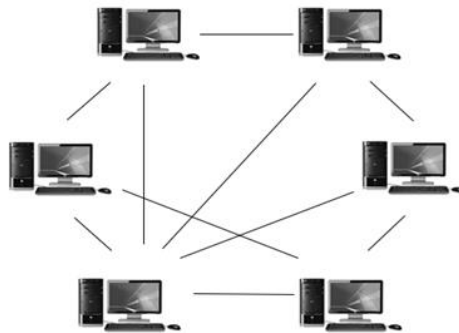
Computer Network Architecture is defined as the physical and logical design of the software, hardware, protocols, and media of the transmission of data. Simply we can say that how computers are organized and how tasks are allocated to the computer.

The two types of network architectures are:

- Peer-To-Peer network
- Client/Server network

Peer-To-Peer network

- Peer-To-Peer network is a network in which all the computers are linked together with equal privilege and responsibilities for processing the data.
- Peer-To-Peer network is useful for small environments, usually up to 10 computers.
- Peer-To-Peer network does not require any dedicated server.
- Every computer on the network can share its resources with any other computer on the network.
- Every computer connected to the network is both client and server.
- Example music file-sharing system and Bit Torrent etc.



Peer-To-Peer network

Advantages of Peer-To-Peer Network:

- It is less costly as it does not contain any dedicated server.
- Failure of one computer may not affect other computers on the network.
- It is easy to set up and maintain as each computer manages itself.

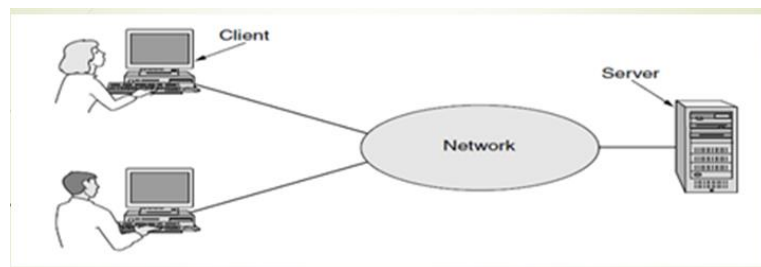
Disadvantages of Peer-To-Peer Network:

- In the case of Peer-To-Peer network, it does not contain the centralized system. So data backup becomes bit difficult here.
- It has a security issue as the device is managed by itself.

Client/ Server Networks

In this model, the data are stored on powerful computers called **servers**. Often these are centrally housed and maintained by a system administrator; In contrast, **clients** are simpler machines. The client and server machines are connected by a network.

- The Client/ server networks involve process, client process and server process, which usually resides on different machines.
- A client process request and receive services from server process.
- A server is responsible for managing all the resources such as files, directories, printer, etc.
- All the clients communicate with each other through a server.



Client/ Server Networks

For Example, a web server could be server process receives request for web pages from clients (Web browser)

Advantages of Client/Server network:

- A Client/Server network contains the centralized system. Therefore, we can back up the data easily.
- A Client/Server network has a dedicated server that improves the overall performance of the whole system.
- Security is better in Client/Server network as a single server administers the shared resources.
- It also increases the speed of the sharing resources.

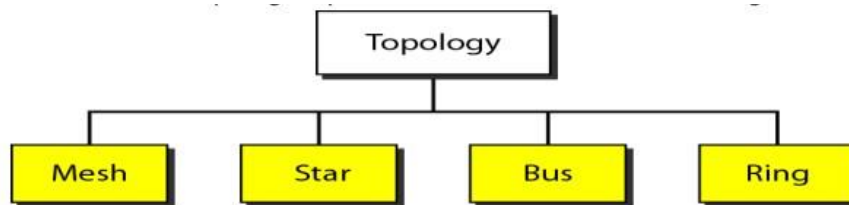
Disadvantages of Client/Server network:

- Client/Server network is expensive as it requires the server with large memory.
- A server has a Network Operating System (NOS) to provide the resources to the clients, but the cost of NOS is very high.

- It requires a dedicated network administrator to manage all the resources.
- If server got crashed entire network will come down and leads to single point of failure.

Physical Topology

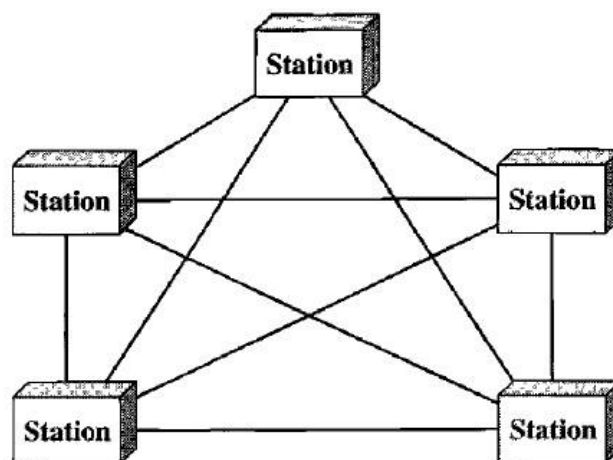
The term physical topology refers to the way in which a network is laid out physically. Two or more devices connect to a link and two or more links form a topology. The topology of a network is the geometric representation of the relationship of all the links and linking devices (usually called nodes) to one another. There are four basic topologies possible: mesh, star, bus, and ring.



1.Mesh Topology

In a mesh topology, every device has a dedicated point-to-point link to every other device. The term dedicated means that the link carries traffic only between the two devices it connects. To find the number of physical links in a fully connected mesh network with n nodes, we first consider that each node must be connected to every other node.

Node 1 must be connected to $n - 1$ nodes, node 2 must be connected to $n - 1$ nodes, and finally node n must be connected to $n - 1$ nodes. We need $n(n - 1)$ physical links. However, if each physical link allows communication in both directions (duplex mode), we can divide the number of links by 2. In other words, we can say that in a mesh topology $n(n - 1) / 2$ links.



A mesh offers several advantages over other network topologies. First, the use of dedicated links guarantees that each connection can carry its own data load, thus eliminating the traffic problems that can occur when links must be shared by multiple devices.

Second, a mesh topology is robust. If one link becomes unusable, it does not incapacitate the entire system.

Third, there is the advantage of privacy or security. When every message travels along a dedicated line, only the intended recipient sees it. Physical boundaries prevent other users from gaining access to messages.

Finally, point-to-point links make fault identification and fault isolation easy. Traffic can be routed to avoid links with suspected problems. This facility enables the network manager to discover the precise location of the fault and aids in finding its cause and solution.

The main disadvantages of a mesh are related to the amount of cabling and the number of I/O ports required. First, because every device must be connected to every other device, installation and reconnection are difficult. Second, the sheer bulk of the wiring can be greater than the available space (in walls, ceilings, or floors) can accommodate.

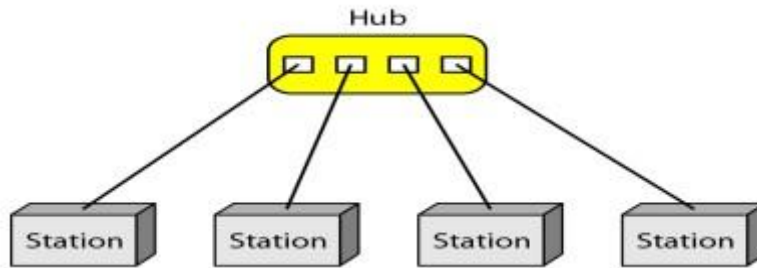
Finally, the hardware required to connect each link (I/O ports and cable) can be more expensive. For these reasons a mesh topology is usually implemented in a limited fashion, for example, as a backbone connecting the main computers of a hybrid network that can include several other topologies.

One practical example of a mesh topology is the connection of telephone regional offices in which each regional office needs to be connected to every other regional office.

2. Star Topology

In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub. The devices are not directly linked to one another. Unlike a mesh topology, a star topology does not allow direct traffic between devices.

The controller acts as an exchange: If one device wants to send data to another, it sends the data to the controller, which then relays the data to the other connected device.



A star topology is less expensive than a mesh topology. In a star, each device needs only one link and one I/O port to connect it to any number of others. This factor also makes it easy to install and reconfigure. Far less cabling needs to be housed, and additions, moves, and deletions involve only one connection between that device and the hub.

Other advantages include robustness. If one link fails, only that link is affected. All other links remain active. This factor also lends itself to easy fault identification and fault isolation. As long as the hub is working, it can be used to monitor link problems and bypass defective links.

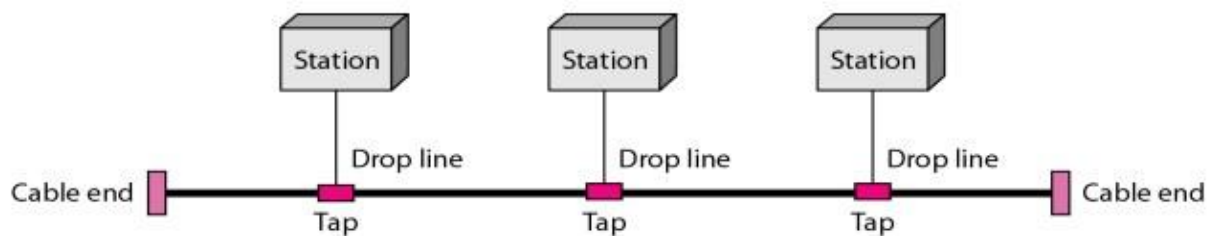
One big disadvantage of a star topology is the dependency of the whole topology on one single point, the hub. If the hub goes down, the whole system is dead.

Although a star requires far less cable than a mesh, each node must be linked to a central hub. For this reason, often more cabling is required in a star than in some other topologies (such as ring or bus).

Example: The star topology is used in local-area networks (LANs).

3. Bus Topology

The preceding examples all describe point-to-point connections. A **bus topology**, on the other hand, is multipoint. One long cable acts as a **backbone** to link all the devices in a network.



Nodes are connected to the bus cable by drop lines and taps. A drop line is a connection running between the device and the main cable. A tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core. As a signal travels along the backbone, some of its energy is transformed into heat. Therefore, it becomes weaker and

weaker as it travels farther and farther. For this reason there is a limit on the number of taps a bus can support and on the distance between those taps.

Advantages of a bus topology include ease of installation. Backbone cable can be laid along the most efficient path, then connected to the nodes by drop lines of various lengths. In this way, a bus uses less cabling than mesh or star topologies. In a star, for example, four network devices in the same room require four lengths of cable reaching all the way to the hub. In a bus, this redundancy is eliminated. Only the backbone cable stretches through the entire facility. Each drop line has to reach only as far as the nearest point on the backbone.

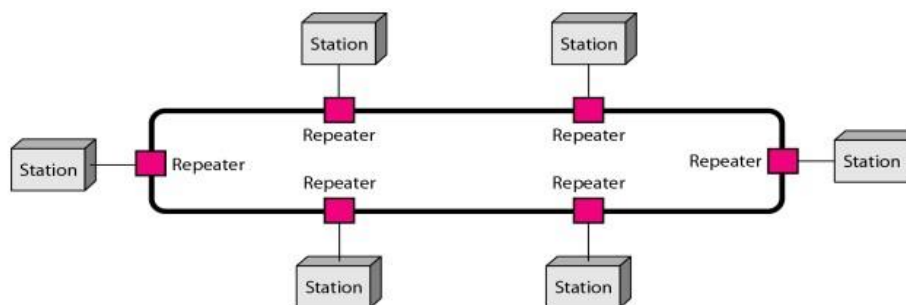
Disadvantages include difficult reconnection and fault isolation. A bus is usually designed to be optimally efficient at installation. It can therefore be difficult to add new devices. Signal reflection at the taps can cause degradation in quality. This degradation can be controlled by limiting the number and spacing of devices connected to a given length of cable. Adding new devices may therefore require modification or replacement of the backbone.

Bus topology was the one of the first topologies used in the design of early local area networks.

Example: Ethernet LANs can use a bus topology, but they are less popular.

4. Ring Topology

In a ring topology, each device has a dedicated point-to-point connection with only the two devices on either side of it. A signal is passed along the ring in one direction, from device to device, until it reaches its destination. Each device in the ring incorporates a repeater. When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along.



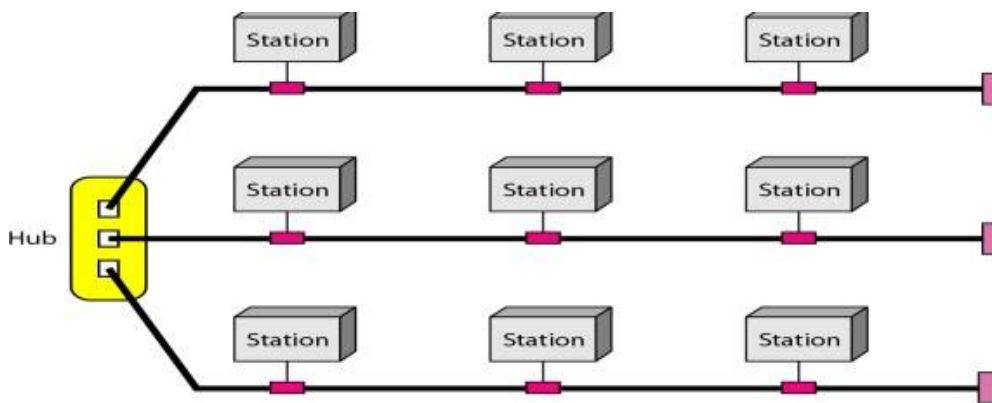
A ring is relatively easy to install and reconfigure. Each device is linked to only its immediate neighbours (either physically or logically). To add or delete a device requires changing only two connections. The only constraints are media and traffic considerations (maximum ring length and number of devices). In addition, fault isolation is simplified.

Generally in a ring, a signal is circulating at all times. If one device does not receive a signal within a specified period, it can issue an alarm. The alarm alerts the network operator to the problem and its location. However, unidirectional traffic can be a disadvantage. In a simple ring, a break in the ring (such as a disabled station) can disable the entire network. This weakness can be solved by using a dual ring or a switch capable of closing off the break.

Ring topology was prevalent when IBM introduced its local-area network Token Ring. Today, the need for higher-speed LANs has made this topology less popular.

5. Hybrid Topology

A network can be hybrid. For example, we can have a main star topology with each branch connecting several stations in a bus topology.



Network Hardware

There is no generally accepted taxonomy into which all computer networks fit, but two dimensions stand out as important: transmission technology and scale.

Broadly speaking, there are two types of transmission technology that are in widespread use: broadcast links and point-to-point links.

Point-to-point links connect individual pairs of machines. To go from the source to the destination on a network made up of point-to-point links, short messages, called packets in certain contexts, may have to first visit one or more intermediate machines. Often multiple routes, of different lengths, are possible, so finding good ones is important in point-to-point networks. Point-to-point transmission with exactly one sender and exactly one receiver is sometimes called unicasting.

In contrast, on a broadcast network, the communication channel is shared by all the machines on the network; packets sent by any machine are received by all the others. An address field within each packet specifies the intended recipient. Upon receiving a packet, a machine checks the address field. If the packet is intended for the receiving machine, that machine processes the packet; if the packet is intended for some other machine, it is just ignored. A wireless network is a common example of a broadcast link, with communication shared over a coverage region that depends on the wireless channel and the transmitting machine. Broadcast systems usually also allow the possibility of addressing a packet to all destinations by using a special code in the address field. When a packet with this code is transmitted, it is received and processed by every machine on the network. This mode of operation is called broadcasting. Some broadcast systems also support transmission to a subset of the machines, which known as multicasting.

An alternative criterion for classifying networks is by scale. Distance is important as a classification metric because different technologies are used at different scales. We classify multiple processor systems by their rough physical size. At the top are the personal area networks, networks that are meant for one person. Beyond these come longer-range networks. These can be divided into local, metropolitan, and wide area networks, each with increasing scale. Finally, the connection of two or more networks is called an internetwork. The worldwide Internet is certainly the best-known (but not the only) example of an internetwork.

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	
100 m	Building	Local area network
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

Categories of Networks

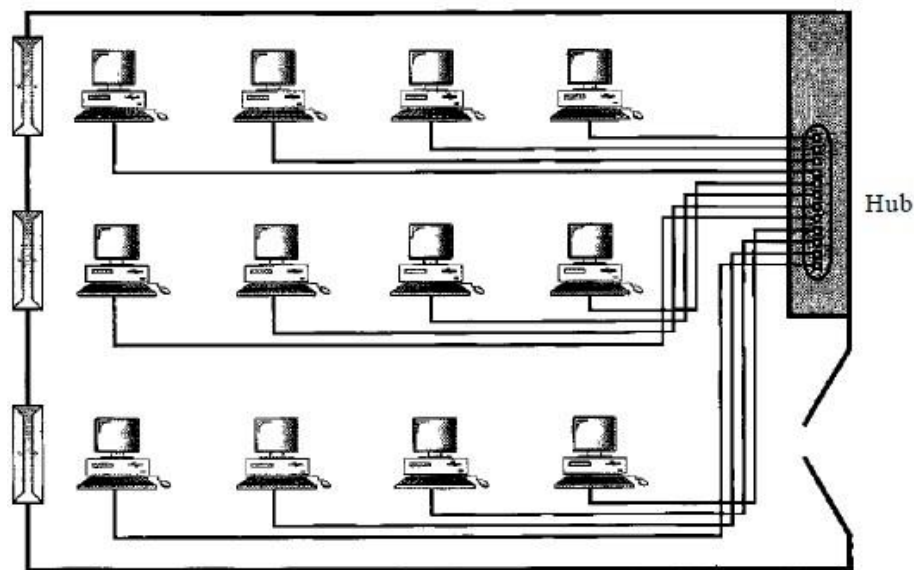
Today when we speak of networks, we are generally referring: Local-area networks, Metropolitan area networks and wide-area networks.

A LAN normally covers an area less than 1Km. A WAN can be worldwide. Networks of a size in between are normally referred to as metropolitan area networks and span tens of miles.

Local Area Network (LAN)

A local area network (LAN) is usually privately owned and links the devices in a single office, building, or campus. Depending on the needs of an organization and the type of technology used, a LAN can be as simple as two PCs and a printer in someone's home office; or it can extend throughout a company and include audio and video peripherals. Currently, LAN size is limited to a few kilometres.

LANs are designed to allow resources to be shared between personal computers or workstations. The resources to be shared can include hardware (e.g., a printer), software (e.g., an application program), or data.



A common example of a LAN, found in many business environments, links a workgroup of task-related computers, for example, engineering workstations or accounting PCs. One of the computers may be given a large capacity disk drive and may become a server to clients. Software can be stored on this central server and used as needed by the whole group. In this example, the size of the LAN may be determined by licensing restrictions on the number of users per copy of software, or by restrictions on the number of users licensed to access the operating system.

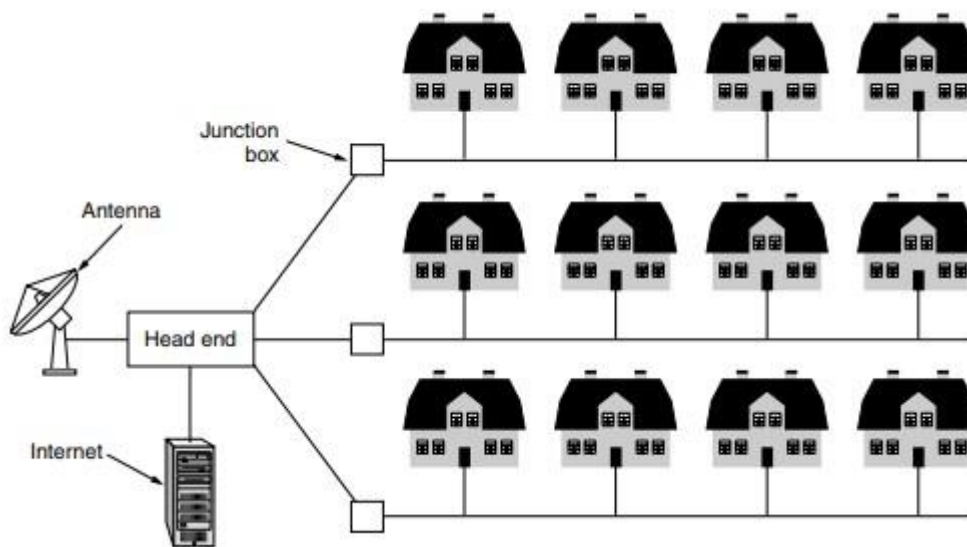
In addition to size, LANs are distinguished from other types of networks by their transmission media and topology. In general, a given LAN will use only one type of transmission medium. The most common LAN topologies are bus, ring, and star. Early LANs had data rates in the 4 to 16 megabits per second (Mbps) range. Today, however, speeds are normally 100 or 1000 Mbps.

Wireless LANs are the newest evolution in LAN technology.

Metropolitan Area Networks (MAN)

A metropolitan area network (MAN) is a network with a size between a LAN and a WAN. It normally covers the area inside a town or a city. It is designed for customers who need a high-speed connectivity, normally to the Internet, and have endpoints spread over a city or part of city. A good example of a MAN is the part of the telephone company network that can provide a high-speed DSL line to the customer.

Another example is the cable TV network that originally was designed for cable TV, but today can also be used for high-speed data connection to the Internet. We discuss DSL lines and cable TV networks.



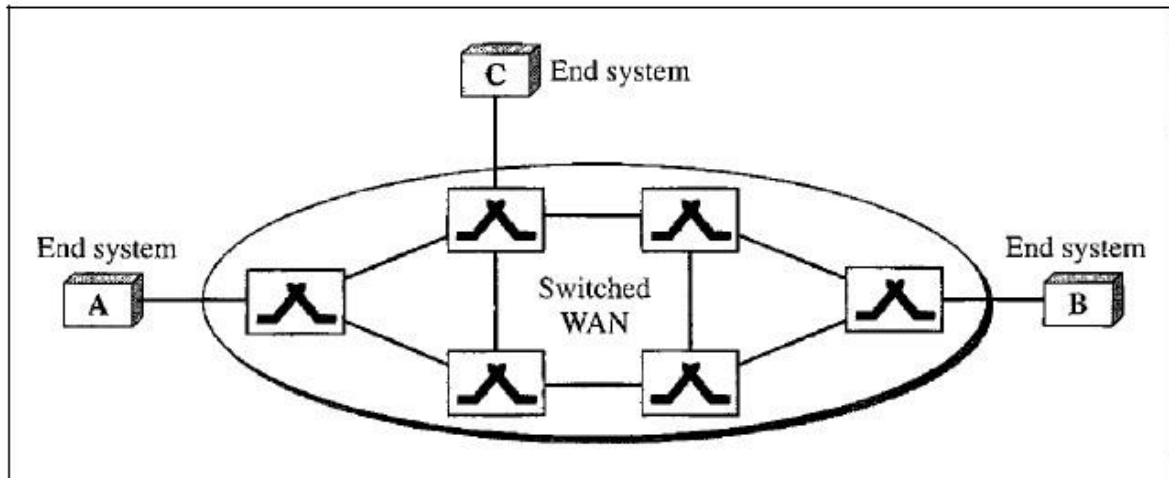
Wide Area Network (WAN)

A wide area network (WAN) provides long-distance transmission of data, image, audio, and video information over large geographic areas that may comprise a country, a continent, or even the whole world. A WAN can be as complex as the backbones that connect the Internet or as simple as a dial-up line that connects a home computer to the Internet. We normally refer to the first as a switched WAN and to the second as a point-to-point WAN.

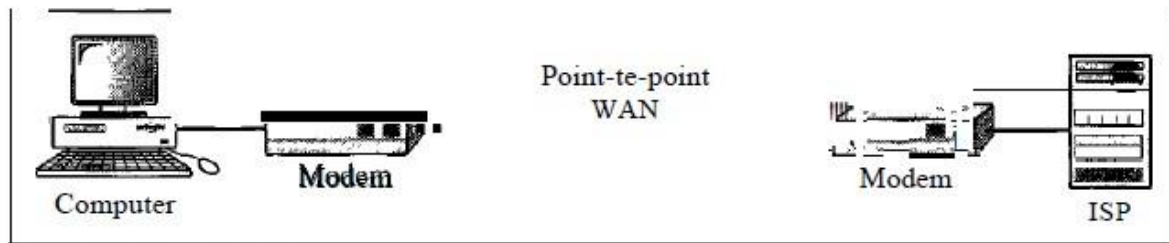
The switched WAN connects the end systems, which usually comprise a router (internetworking connecting device) that connects to another LAN or WAN.

The point-to-point WAN is normally a line leased from a telephone or cable TV provider that connects a home computer or a small LAN to an Internet service provider (ISP).

This type of WAN is often used to provide Internet access.



a. Switched WAN



b. Point-to-point WAN

Interconnection of Networks: Internetwork

Today, it is very rare to see a LAN, a MAN, or a LAN in isolation; they are connected to one another. When two or more networks are connected, they become an internetwork, or internet.

As an example, assume that an organization has two offices, one on the east coast and the other on the west coast. The established office on the west coast has a bus topology LAN; the newly opened office on the east coast has a star topology LAN.

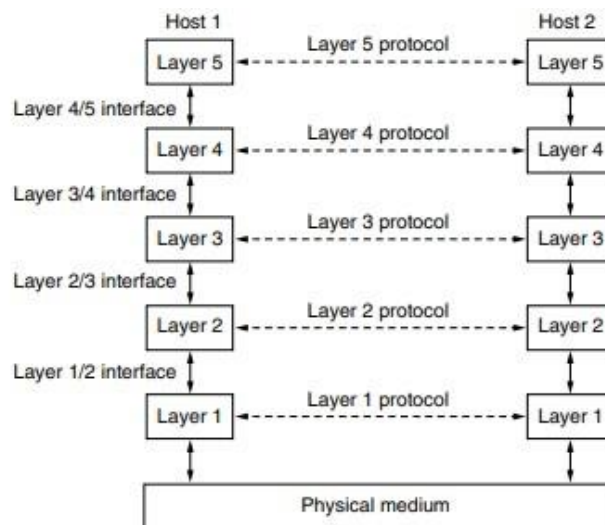
To create a backbone WAN for connecting these three entities (two LANs and the president's computer), a switched WAN (operated by a service provider such as a telecom company) has been leased. To connect the LANs to this switched WAN, however, three point-to-point WANs are required. These point-to-point WANs can be a high-speed DSL line offered by a telephone company or a cable modern line offered by a cable TV provider.

Network Software: Protocol Hierarchies:

To reduce their design complexity, most networks are organized as a stack of layers or levels, each one built upon the one below it. The number of layers, the name of each layer, the contents of each layer, and the function of each layer differ from network to network. The purpose of each layer is to offer certain services to the higher layers while shielding those layers from the details of how the offered services are actually implemented. In a sense, each layer is a kind of virtual machine, offering certain services to the layer above it.

When layer 'n' on one machine carries on a conversation with layer 'n' on another machine, the rules and conventions used in this conversation are collectively known as the layer n protocol. Basically, a protocol is an agreement between the communicating parties on how communication is to proceed.

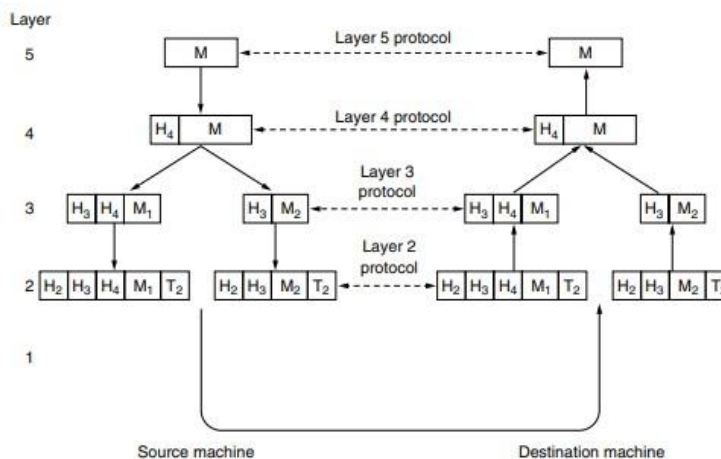
A five-layer network is illustrated in below Fig. The entities comprising the corresponding layers on different machines are called peers. The peers may be software processes, hardware devices, or even human beings. In other words, it is the peers that communicate by using the protocol to talk to each other.



In reality, no data are directly transferred from layer n on one machine to layer n on another machine. Instead, each layer passes data and control information to the layer immediately below it, until the lowest layer is reached. Below layer 1 is the physical medium through which actual communication occurs. Between each pair of adjacent layers is an interface. The interface defines which primitive operations and services the lower layer makes available to the upper one.

A set of layers and protocols is called a network architecture. A list of the protocols used by a certain system, one protocol per layer, is called a protocol stack.

Now consider a more technical example: how to provide communication to the top layer of the five-layer network in below Fig. A message, M, is produced by an application process running in layer 5 and given to layer 4 for transmission. Layer 4 puts a header in front of the message to identify the message and passes the result to layer 3. The header includes control information, such as addresses, to allow layer 4 on the destination machine to deliver the message. Other examples of control information used in some layers are sequence numbers (in case the lower layer does not preserve message order), sizes, and times. In many networks, no limit is placed on the size of messages transmitted in the layer 4 protocol but there is nearly always a limit imposed by the layer 3 protocol. Consequently, layer 3 must break up the incoming messages into smaller units, packets, prepending a layer 3 header to each packet. In this example, M is split into two parts, M₁ and M₂, that will be transmitted separately. Layer 3 decides which of the outgoing lines to use and passes the packets to layer 2. Layer 2 adds to each piece not only a header but also a trailer, and gives the resulting unit to layer 1 for physical transmission. At the receiving machine the message moves upward, from layer to layer, with headers being stripped off as it progresses. None of the headers for layers below n are passed up to layer n.



Design Issues for the Layers:

Error Control

Flow control

Multiplexing/Demultiplexing

Routing

Connection-oriented and Connectionless Services:

Connection-oriented service is modelled after the telephone system. To talk to someone, you pick up the phone, dial the number, talk, and then hang up. Similarly, to use a connection-oriented network service, the service user first establishes a connection, uses the connection, and then releases the connection.

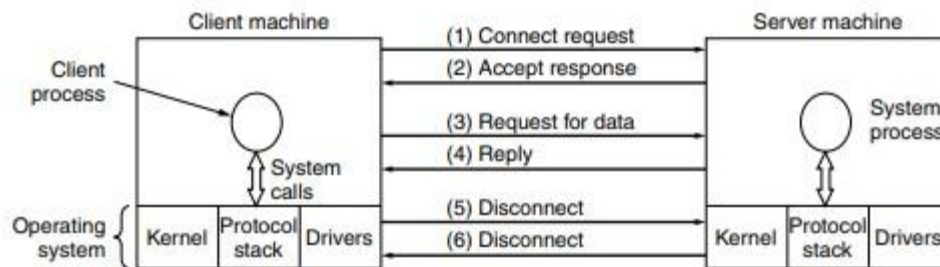
In contrast to connection-oriented service, connectionless service is modelled after the postal system. Each message (letter) carries the full destination address, and each one is routed through the intermediate nodes inside the system independent of all the subsequent messages.

	Service	Example
Connection-oriented	Reliable message stream	Sequence of pages
	Reliable byte stream	Movie download
	Unreliable connection	Voice over IP
Connection-less	Unreliable datagram	Electronic junk mail
	Acknowledged datagram	Text messaging
	Request-reply	Database query

Service Primitives:

A service is formally specified by a set of primitives (operations) available to user processes to access the service. These primitives tell the service to perform some action or report on an action taken by a peer entity. If the protocol stack is located in the operating system, as it often is, the primitives are normally system calls. These calls cause a trap to kernel mode, which then turns control of the machine over to the operating system to send the necessary packets. The set of primitives available depends on the nature of the service being provided. The primitives for connection-oriented service are different from those of connectionless service. As a minimal example of the service primitives that might provide a reliable byte stream, consider the primitives listed

Primitive	Meaning
LISTEN	Block waiting for an incoming connection
CONNECT	Establish a connection with a waiting peer
ACCEPT	Accept an incoming connection from a peer
RECEIVE	Block waiting for an incoming message
SEND	Send a message to the peer
DISCONNECT	Terminate a connection



Reference Models:

THE OSI MODEL:

The ISO Model has established in 1947, the International Standards Organization (ISO) is a multinational body dedicated to worldwide agreement on international standards. An ISO standard that covers all aspects of network communications is the Open Systems Interconnection model. It was first introduced in the late 1970s. An open system is a set of protocols that allows any two different systems to communicate regardless of their underlying architecture.

The purpose of the OSI model is to show how to facilitate communication between different systems without requiring changes to the logic of the underlying hardware and software. The OSI model is not a protocol; it is a model for understanding and designing a network architecture that is flexible, robust, and interoperable.

The OSI model is a layered framework for the design of network systems that allows communication between all types of computer systems. It consists of seven separate but related layers, each of which defines a part of the process of moving information across a network.

The layers are:

1. Physical Layer
2. Data Link Layer
3. Network Layer
4. Transport Layer

5. Session Layer

6. Presentation Layer

7. Application Layer

Layered Architecture:

The OSI model is composed of seven ordered layers: physical (layer 1), data link (layer 2), network (layer 3), transport (layer 4), session (layer 5), presentation (layer 6), and application (layer 7). The below figure shows the layers involved when a message is sent from device A to device B. As the message travels from A to B, it may pass through many intermediate nodes. These intermediate nodes usually involve only the first three layers of the OSI model.

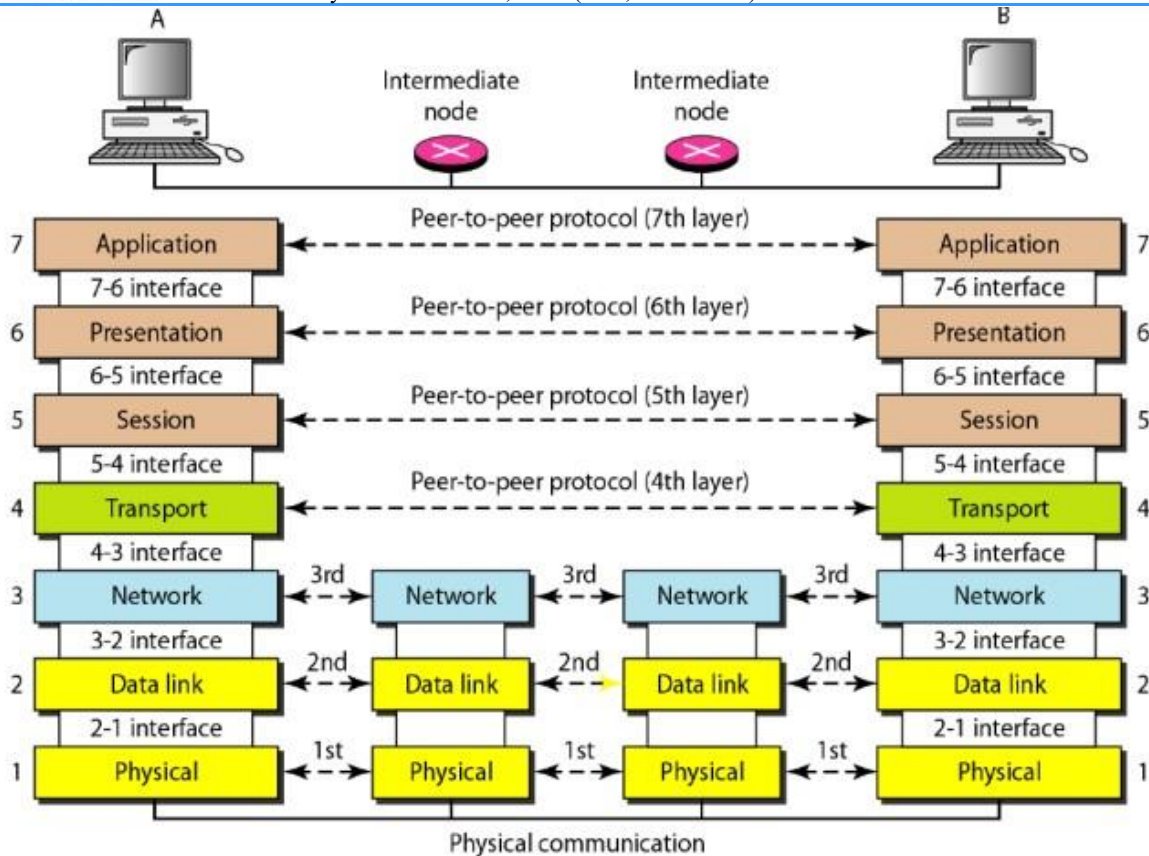
In *developing* the model, the designers distilled the process of transmitting data to its most fundamental elements. They identified which networking functions had related uses and collected those functions into discrete groups that became the layers. Each layer defines a family of functions distinct from those of the other layers. By defining and localizing functionality in this fashion, the designers created an architecture that is both comprehensive and flexible. Most importantly, the OSI model allows complete interoperability between otherwise incompatible systems.

Within a single machine, each layer calls upon the services of the layer just below it. Layer 3, for example, uses the services provided by layer 2 and provides services for layer 4. Between machines, layer x on one machine communicates with layer x on another machine. This communication is governed by an agreed-upon series of rules and conventions called protocols. The processes on each machine that communicate at a given layer are called peer-to-peer processes. Communication between machines is therefore a peer-to-peer process using the protocols appropriate to a given layer.

Peer-to-Peer Processes

At the physical layer, communication is direct: In Figure, device A sends a stream of bits to device B (through intermediate nodes). At the higher layers, however, communication must move down through the layers on device A, over to device B, and then back up through the layers. Each layer in the sending device adds its own information to the message it receives from the layer just above it and passes the whole package to the layer just below it.

At layer 1 the entire package is converted to a form that can be transmitted to the receiving device. At the receiving machine, the message is unwrapped layer by layer, with each process receiving and removing the data meant for it. For example, layer 2 removes the data meant for it, then passes the rest to layer 3. Layer 3 then removes the data meant for it and passes the rest to layer 4, and so on.



Interfaces between Layers

The passing of the data and network information down through the layers of the sending device and back up through the layers of the receiving device is made possible by an interface between each pair of adjacent layers.

Each interface defines the information and services a layer must provide for the layer above it. Well-defined interfaces and layer functions provide modularity to a network. As long as a layer provides the expected services to the layer above it, the specific implementation of its functions can be modified or replaced without requiring changes to the surrounding layers.

Organization of the Layers

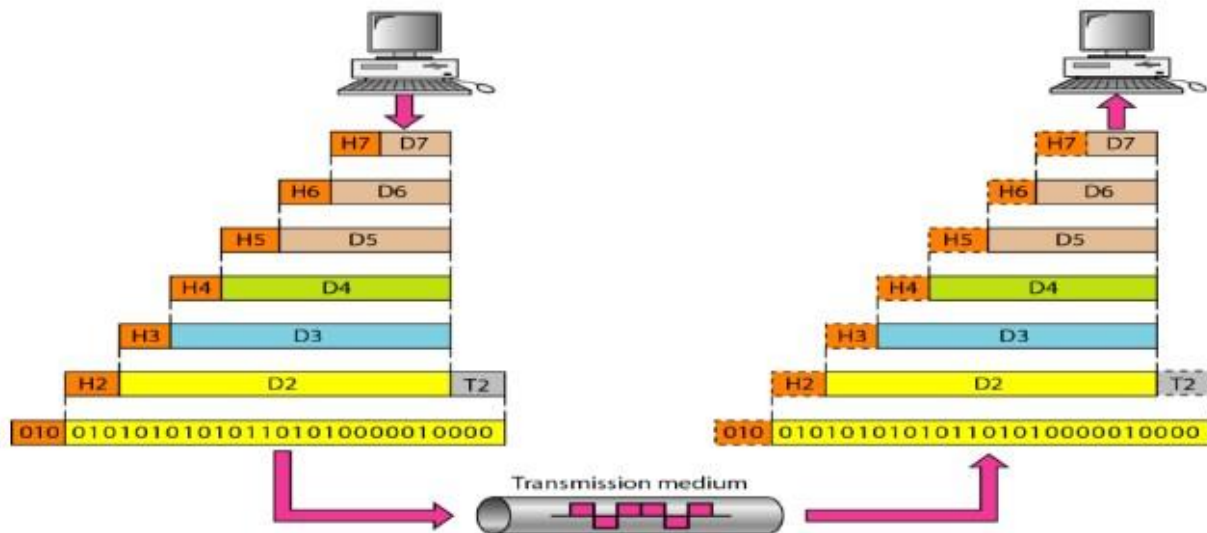
The seven layers can be thought of as belonging to three subgroups. Layers 1, 2, and 3 physical, data link, and network are the network support layers; they deal with the physical aspects of moving data from one device to another (such as electrical specifications, physical connections, physical addressing, and transport timing and reliability).

Layers 5, 6, and 7-session, presentation, and application-can be thought of as the user support layers; they allow interoperability among unrelated software systems. Layer 4, the transport layer,

links the two subgroups and ensures that what the lower layers have transmitted is in a form that the upper layers can use.

The upper OSI layers are almost always implemented in software; lower layers are a combination of hardware and software, except for the physical layer, which is mostly hardware. The below figure gives an overall view of the OSI layers, D7 means the data unit at layer 7, D6 means the data unit at layer 6, and so on. The process starts at layer 7 (the application layer), then moves from layer to layer in descending, sequential order. At each layer, a **header**, or possibly a **trailer**, can be added to the data unit.

Commonly, the trailer is added only at layer 2. When the formatted data unit passes through the physical layer (layer 1), it is changed into an electromagnetic signal and transported along a physical link.



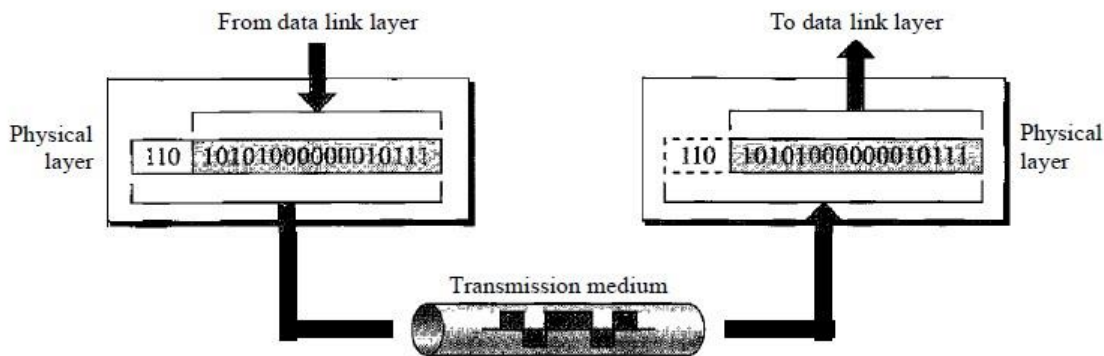
Upon reaching its destination, the signal passes into layer 1 and is transformed back into digital form. The data units then move back up through the OSI layers. As each block of data reaches the next higher layer, the headers and trailers attached to it at the corresponding sending layer are removed, and actions appropriate to that layer are taken. By the time it reaches layer 7, the message is again in a form appropriate to the application and is made available to the recipient.

LAYERS IN THE OSI MODEL:

1. Physical Layer

The physical layer coordinates the functions required to carry a bit stream over a physical medium. It deals with the mechanical and electrical specifications of the interface and transmission medium.

It also defines the procedures and functions that physical devices and interfaces have to perform for transmission to Occur.



The physical layer is responsible for movements of individual bits from one hop (node) to the next.

The physical layer is also concerned with the following:

Physical characteristics of interfaces and medium.

The physical layer defines the characteristics of the interface between the devices and the transmission medium. It also defines the type of transmission medium.

Representation of bits

The physical layer data consists of a stream of bits (sequence of 0's or 1's) with no interpretation. To be transmitted, bits must be encoded into signals-electrical or optical. The physical layer defines the type of encoding (how 0's and 1's are changed to signals).

Data rate

The transmission rate-the number of bits sent each second-is also defined by the physical layer. In other words, the physical layer defines the duration of a bit, which is how long it lasts.

Synchronization of bits

The sender and receiver not only must use the same bit rate but also must be synchronized at the bit level. In other words, the sender and the receiver clocks must be synchronized.

Line configuration

The physical layer is concerned with the connection of devices to the media. In a point-to-point configuration, two devices are connected through a dedicated link. In a multipoint configuration, a link is shared among several devices.

Physical topology

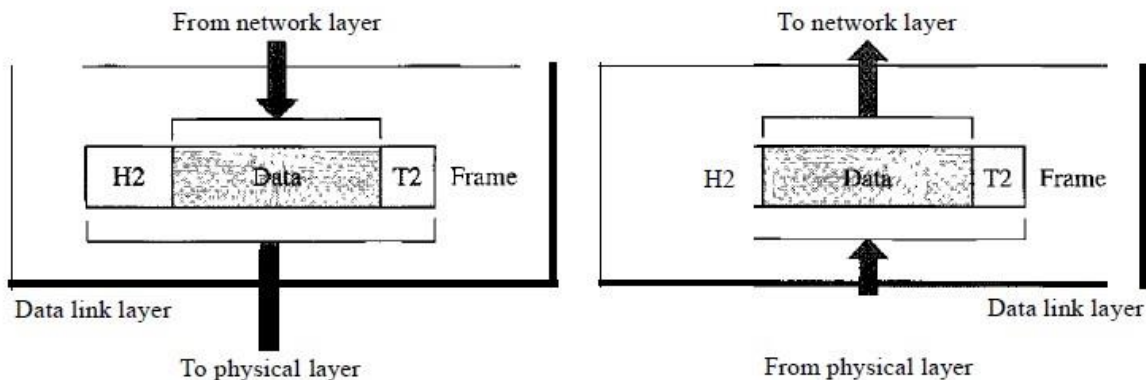
The physical topology defines how devices are connected to make a network. Devices can be connected by using a mesh topology (every device is connected to every other device), a star topology (devices are connected through a central device), a ring topology (each device is connected to the next, forming a ring), a bus topology (every device is on a common link), or a hybrid topology (this is a combination of two or more topologies).

Transmission mode

The physical layer also defines the direction of transmission between two devices: simplex, half-duplex, or full-duplex. In simplex mode, only one device can send; the other can only receive. The simplex mode is a one-way communication. In the half-duplex mode, two devices can send and receive, but not at the same time. In a full-duplex (or simply duplex) mode, two devices can send and receive at the same time.

2. Data Link Layer

The data link layer transforms the physical layer, a raw transmission facility, to a reliable link. It makes the physical layer appear error-free to the upper layer (network layer). The data link layer is responsible for moving frames from one hop (node) to the next.



Framing

The data link layer divides the stream of bits received from the network layer into manageable data units called frames.

Physical addressing

If frames are to be distributed to different systems on the network, the data link layer adds a header to the frame to define the sender and/or receiver of the frame. If the frame is intended for a system outside the sender's network, the receiver address is the address of the device that connects the network to the next one.

Flow control

If the rate at which the data are absorbed by the receiver is less than the rate at which data are produced in the sender, the data link layer imposes a flow control mechanism to avoid overwhelming the receiver.

Error control

The data link layer adds reliability to the physical layer by adding mechanisms to detect and retransmit damaged or lost frames. It also uses a mechanism to recognize duplicate frames. Error control is normally achieved through a trailer added to the end of the frame.

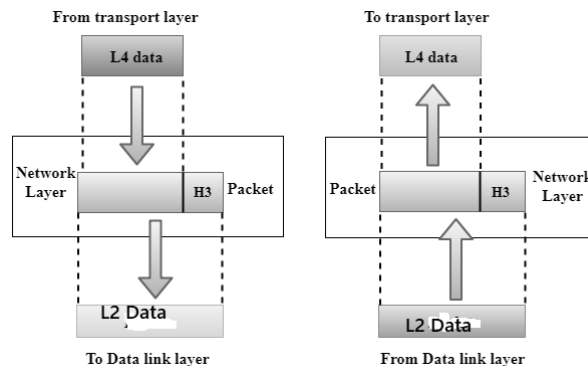
Access control

When two or more devices are connected to the same link, data link layer protocols are necessary to determine which device has control over the link at any given time.

3. Network Layer

The network layer is responsible for the source-to-destination delivery of a packet, possibly across multiple networks (links). Whereas the data link layer oversees the delivery of the packet between two systems on the same network (links), the network layer ensures that each packet gets from its point of origin to its final destination.

The network layer is responsible for the delivery of individual packets from the source host to the destination host.

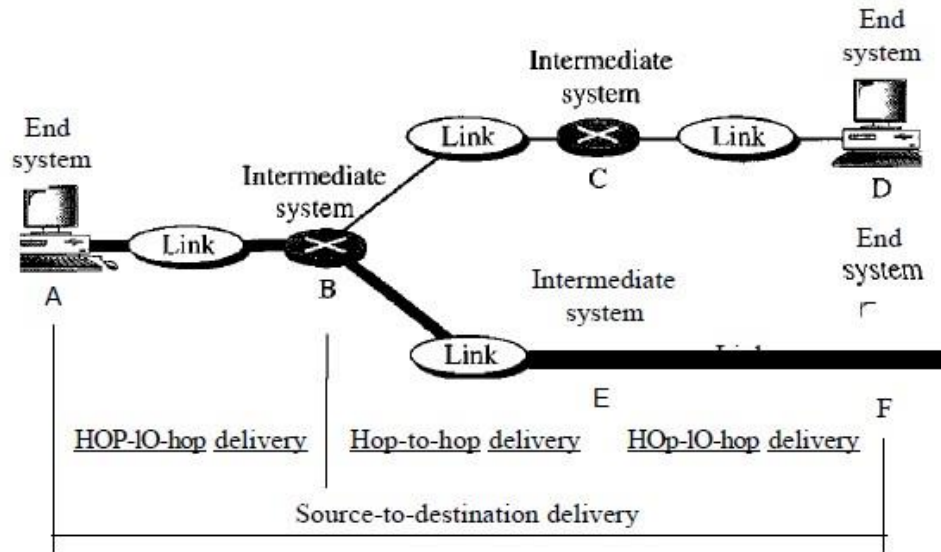


Logical addressing

The physical addressing implemented by the data link layer handles the addressing problem locally. If a packet passes the network boundary, we need another addressing system to help distinguish the source and destination systems. The network layer adds a header to the packet coming from the upper layer that, among other things, includes the logical addresses of the sender and receiver.

Routing

When independent networks or links are connected to create *internetworks* (network of networks) or a large network, the connecting devices (called *routers* or *switches*) route or switch the packets to their final destination. One of the functions of the network layer is to provide this mechanism.

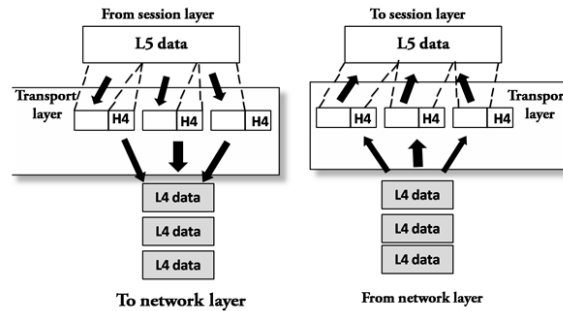


The network layer at A sends the packet to the network layer at B. When the packet arrives at router B, the router makes a decision based on the final destination (F) of the packet. The router B uses its routing table to find that the next hop is router E. The network layer at B, therefore, sends the packet to the network layer at E. The network layer at E, in turn, sends the packet to the network layer at F.

4. Transport Layer

The transport layer is responsible for process-to-process delivery of the entire message. A process is an application program running on a host. Whereas the network layer oversees source to-destination delivery of individual packets, it does not recognize any relationship between those packets. It treats each one independently, as though each piece belonged to a separate message, whether or not it does. The transport layer, on the other hand, ensures that the whole message arrives intact and in order, overseeing both error control and flow control at the source to-destination level.

The transport layer is responsible for the delivery of a message from one process to another.



Service-point addressing

Computers often run several programs at the same time. For this reason, source-to-destination delivery means delivery not only from one computer to the next but also from a specific process (running program) on one computer to a specific process (running program) on the other. The transport layer header must therefore include a type of address called a *service-point address* (or port address). The network layer gets each packet to the correct computer; the transport layer gets the entire message to the correct process on that computer.

Segmentation and reassembly

A message is divided into transmittable segments, with each segment containing a sequence number. These numbers enable the transport layer to reassemble the message correctly upon arriving at the destination and to identify and replace packets that were lost in transmission.

Connection control

The transport layer can be either connectionless or connection oriented. A connectionless transport layer treats each segment as an independent packet and delivers it to the transport layer at the destination machine. A connection oriented transport layer makes a connection with the transport layer at the destination machine first before delivering the packets. After all the data are transferred, the connection is terminated.

Flow control

Like the data link layer, the transport layer is responsible for flow control. However, flow control at this layer is performed end to end rather than across a single link.

Error control

Like the data link layer, the transport layer is responsible for error control. However, error control at this layer is performed process-to process rather than across a single link. The sending transport layer makes sure that the entire message arrives at the receiving transport layer without error (damage, loss, or duplication). Error correction is usually achieved through retransmission.

5. Session Layer

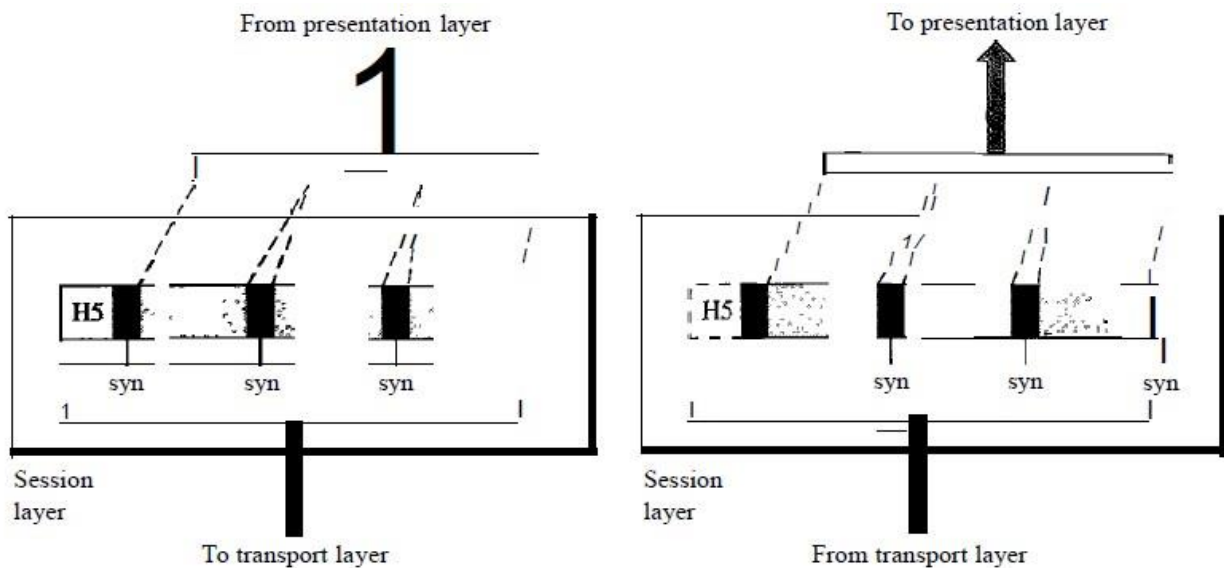
The services provided by the first three layers (physical, data link, and network) are not sufficient for some processes. The session layer is the network *dialog controller*. It establishes, maintains, and synchronizes the interaction among communicating systems. The session layer is responsible for dialog control and synchronization.

Dialog control

The session layer allows two systems to enter into a dialog. It allows the communication between two processes to take place in either half duplex (one way at a time) or full-duplex (two ways at a time) mode.

Synchronization

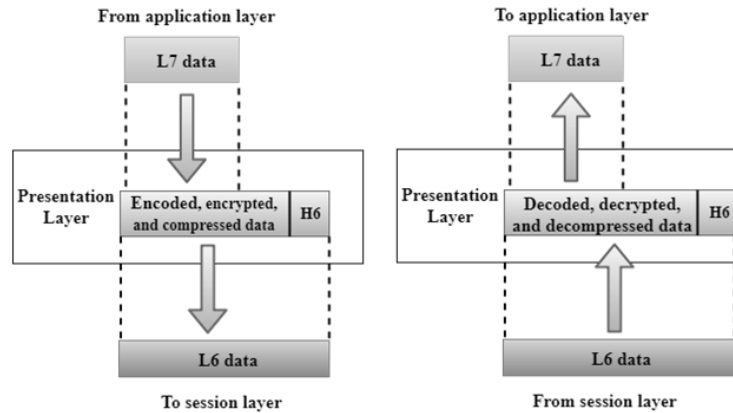
The session layer allows a process to add checkpoints, or synchronization points, to a stream of data. For example, if a system is sending a file of 2000 pages, it is advisable to insert checkpoints after every 100 pages to ensure that each 100-page unit is received and acknowledged independently. In this case, if a crash happens during the transmission of page 523, the only pages that need to be resent after system recovery are pages 501 to 523. Pages previous to 501 need not be resent.



6. Presentation Layer

The presentation layer is concerned with the syntax and semantics of the information exchanged between two systems.

The presentation layer is responsible for translation, compression, and encryption.



Translation

The processes (running programs) in two systems are usually exchanging information in the form of character strings, numbers, and so on. The information must be changed to bit streams before being transmitted. Because different computers use different encoding systems, the presentation layer is responsible for interoperability between these different encoding methods.

The presentation layer at the sender changes the information from its sender-dependent format into a common format. The presentation layer at the receiving machine changes the common format into its receiver-dependent format.

Encryption

To carry sensitive information, a system must be able to ensure privacy. Encryption means that the sender transforms the original information to another form and sends the resulting message out over the network. Decryption reverses the original process to transform the message back to its original form.

Compression

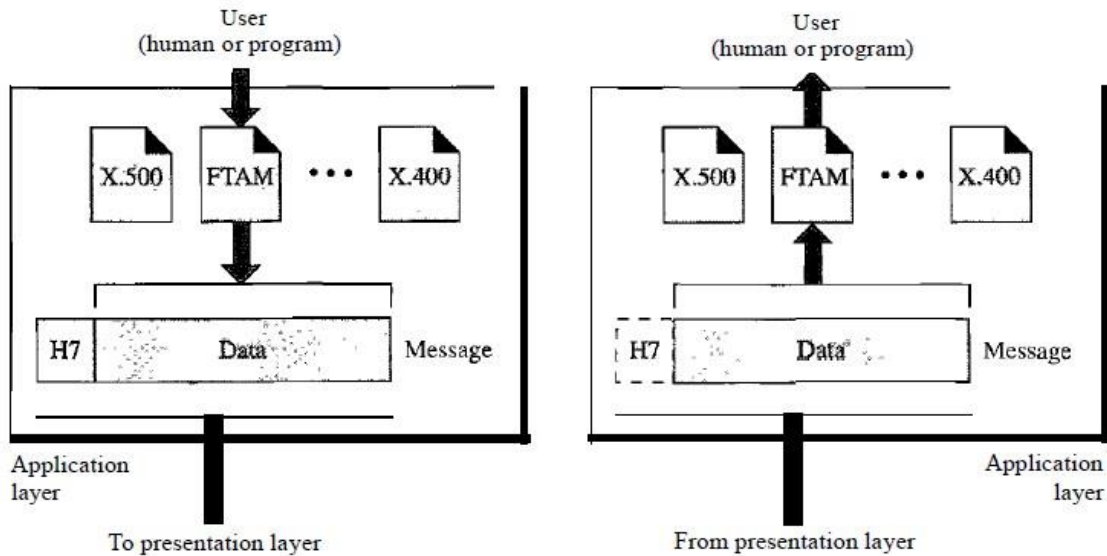
Data compression reduces the number of bits contained in the information. Data compression becomes particularly important in the transmission of multimedia such as text, audio, and video.

7. Application Layer

The application layer enables the user, whether human or software, to access the network. It provides user interfaces and support for services such as electronic mail, remote file access and transfer, shared database management, and other types of distributed information services. Figure shows the relationship of the application layer to the user and the presentation layer. Of the many application services available, the figure shows only three: X.400 (message-handling services),

X.500 (directory services), and file transfer, access, and management (FTAM). The user in this example employs X.400 to send an e-mail message.

The application layer is responsible for providing services to the user.



Network virtual terminal

A network virtual terminal is a software version of a physical terminal, and it allows a user to log on to a remote host. To do so, the application creates a software emulation of a terminal at the remote host. The user's computer talks to the software terminal which, in turn, talks to the host, and vice versa. The remote host believes it is communicating with one of its own terminals and allows the user to log on.

File transfer, access, and management

This application allows a user to access files in a remote host (to make changes or read data), to retrieve files from a remote computer for use in the local computer, and to manage or control files in a remote computer locally.

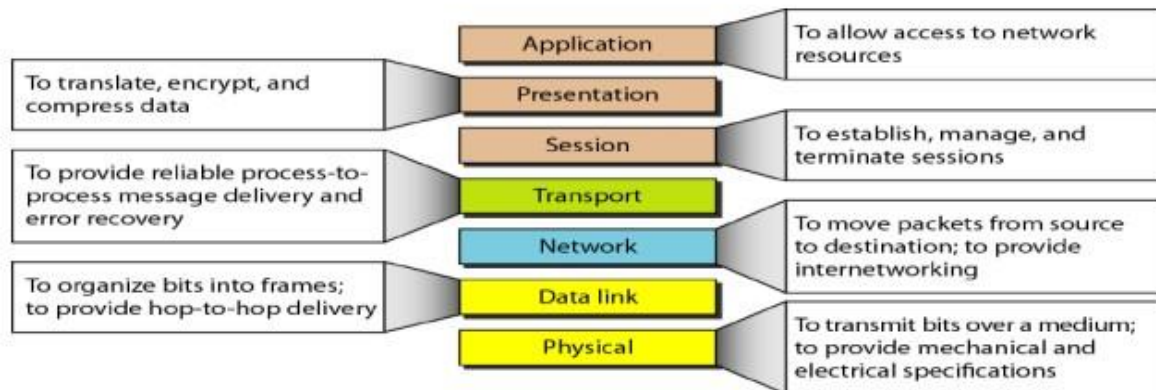
Mail services

This application provides the basis for e-mail forwarding and storage.

Directory services

This application provides distributed database sources and access for global information about various objects and services.

Summary of layers



TCP/IP REFERENCE MODEL (PROTOCOL SUITE):

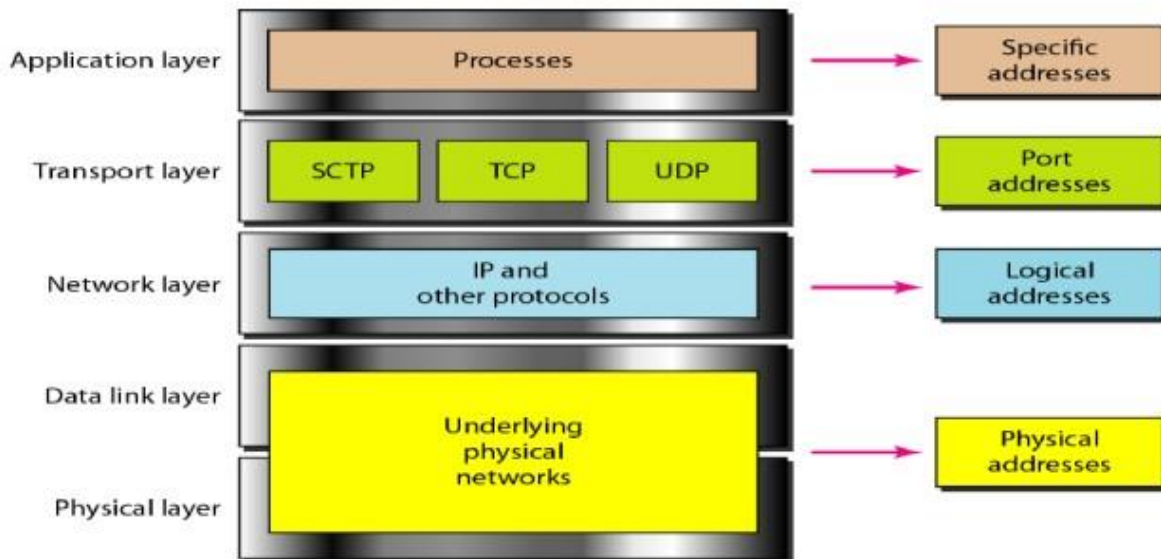
The TCP/IP protocol suite was developed prior to the OSI model. Therefore, the layers in the TCP/IP protocol suite do not exactly match those in the OSI model. The original TCP/IP protocol suite was defined as having four layers: host-to-network, internet, transport, and application. However, when TCP/IP is compared to OSI, we can say that the host-to-network layer is equivalent to the combination of the physical and data link layers.

The internet layer is equivalent to the network layer, and the application layer is roughly doing the job of the session, presentation, and application layers with the transport layer in TCP/IP taking care of part of the duties of the session layer.

The TCP/IP protocol suite is made of five layers:

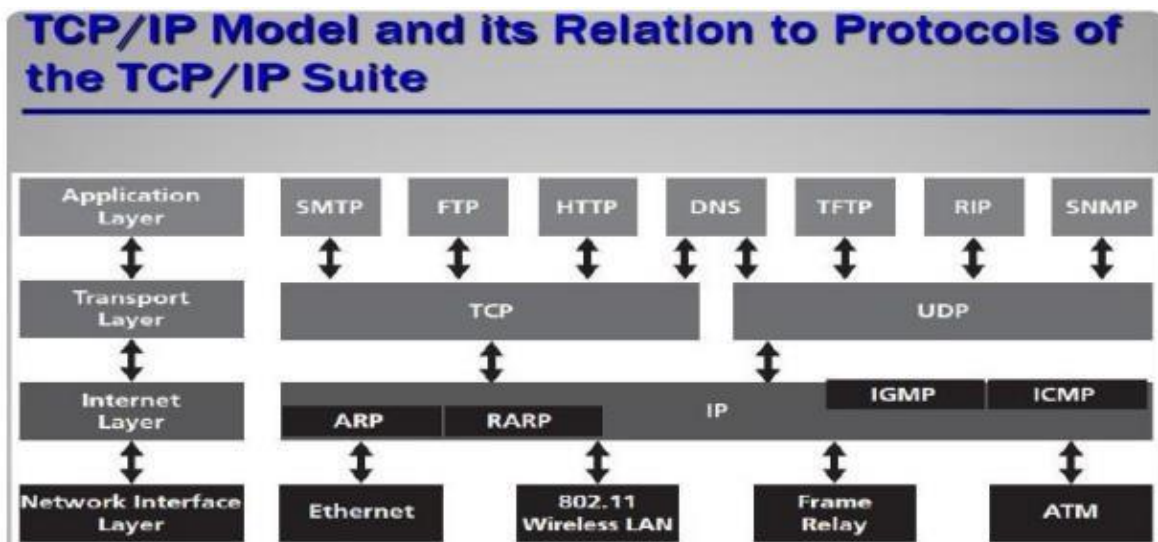
1. Physical Layer
2. Data link Layer
3. Network Layer
4. Transport Layer
5. Application Layer

The first four layers provide physical standards, network interfaces, internetworking, and transport functions that correspond to the first four layers of the OSI model. The three topmost layers in the OSI model, however, are represented in TCP/IP by a single layer called the *application layer*.



TCP/IP is a hierarchical protocol made up of interactive modules, each of which provides a specific functionality; however, the modules are not necessarily interdependent. Whereas the OSI model specifies which functions belong to each of its layers, the layers of the *TCP/IP* protocol suite contain relatively independent protocols that can be mixed and matched depending on the needs of the system. The term *hierarchical* means that each upper-level protocol is supported by one or more lower-level protocols.

At the transport layer, *TCP/IP* defines three protocols: Transmission Control Protocol (TCP), User Datagram Protocol (UDP), and Stream Control Transmission Protocol (SCTP). At the network layer, the main protocol defined by TCP/IP is the Internetworking Protocol (IP); there are also some other protocols that support data movement in this layer.



1. Physical and Data Link Layers (HOST-to-Network Layer)

At the physical and data link layers, TCP/IP does not define any specific protocol. It supports all the standard and proprietary protocols. A network in a TCP/IP internetwork can be a local-area network or a wide-area network.

2. Network Layer (Internet Layer)

At the network layer (or, more accurately, the internetwork layer), *TCP/IP* supports the Internetworking Protocol. IP, in turn, uses four supporting protocols: ARP, RARP, ICMP, and IGMP.

Internetworking Protocol (IP)

The Internetworking Protocol (IP) is the transmission mechanism used by the TCP/IP protocols. It is an unreliable and connectionless protocol—a best-effort delivery service.

The term *best effort* means that IP provides no error checking or tracking. IP assumes the unreliability of the underlying layers and does its best to get a transmission through to its destination, but with no guarantees.

IP transports data in packets called *datagram*'s, each of which is transported separately. Datagram's can travel along different routes and can arrive out of sequence or be duplicated. IP does not keep track of the routes and has no facility for reordering datagrams once they arrive at their destination.

Address Resolution Protocol

The Address Resolution Protocol (ARP) is used to associate a logical address with a physical address. On a typical physical network, such as a LAN, each device on a link is identified by a physical or station address, usually imprinted on the network interface card (NIC).

ARP is used to find the physical address of the node when its Internet address is known.

Reverse Address Resolution Protocol

The Reverse Address Resolution Protocol (RARP) allows a host to discover its Internet address when it knows only its physical address. It is used when a computer is connected to a network for the first time or when a diskless computer is booted.

Internet Control Message Protocol

The Internet Control Message Protocol (ICMP) is a mechanism used by hosts and gateways to send notification of datagram problems back to the sender. ICMP sends query and error reporting messages.

Internet Group Message Protocol

The Internet Group Message Protocol (IGMP) is used to facilitate the simultaneous transmission of a message to a group of recipients.

3. Transport Layer

Traditionally the transport layer was represented in *TCP/IP* by two protocols: TCP and UDP. IP is a host-to-host protocol, meaning that it can deliver a packet from one physical device to another. UDP and TCP are transport level protocols responsible for delivery of a message from a process (running program) to another process. A new transport layer protocol, SCTP, has been devised to meet the needs of some newer applications.

User Datagram Protocol

The User Datagram Protocol (UDP) is the simpler of the two standard TCP/IP transport protocols. It is a process-to-process protocol that adds only port addresses, checksum error control, and length information to the data from the upper layer.

Transmission Control Protocol

The Transmission Control Protocol (TCP) provides full transport-layer services to applications. TCP is a reliable stream transport protocol. The term *stream*, in this context, means connection-oriented: A connection must be established between both ends of a transmission before either can transmit data.

At the sending end of each transmission, TCP divides a stream of data into smaller units called *segments*. Each segment includes a sequence number for reordering after receipt, together with an acknowledgment number for the segments received. Segments are carried across the internet inside of IP datagram's. At the receiving end, TCP collects each datagram as it comes in and reorders the transmission based on sequence numbers.

Stream Control Transmission Protocol

The Stream Control Transmission Protocol (SCTP) provides support for newer applications such as voice over the Internet. It is a transport layer protocol that combines the best features of UDP and TCP.

4. Application Layer

The *application layer* in TCP/IP is equivalent to the combined session, presentation, and application layers in the OSI model many protocols are defined at this layer.

Comparison of OSI and TCP/IP Reference Model:

The OSI and TCP/IP reference models have much in common. Both are based on the concept of a stack of independent protocols.

The following differences show the importance of OSI and TCP/IP reference model.

OSI model	TCP/IP Reference model
1. OSI is a generic, protocol independent standard, acting as a communication gateway between the n/w and end user.	1. TCP/IP model is based on standard protocols around which the internet has developed. It is a communication protocol which allows connection of host overall n/w.
2. In OSI model the transport layer guarantees the delivery of packets	2. In TCP/IP model the transport layer does not guarantees delivery of packets. It is more reliable.
3. It follows vertical approach.	3. Follows horizontal approach.
4. OSI has a separate presentation layer and session layer	4. It does not have a presentation and layer.
5. OSI is a reference model around which the networks are built.	5. TCP/IP model is in a way implementation of model.
6. Network layer of OSI model provides both connection oriented and connection less services.	6. The n/w layer in TCP/IP model provides connection less service
7. OSI model has a problem of fitting the protocols into the model	7. TCP/IP model does not fit any protocol
8. Protocol are hidden in OSI model and easily replaced as the technology changes.	8. In TCP/IP replacing protocol is not easy
9. OSI model defines services interface and protocols very clearly and makes clear distinction between them	9. In TCP/IP, service, Interface and protocols are not clearly separated.

10. It is protocol independent
11. It has 7 layers
12. The layers are
 1. Physical
 2. Data link
 3. Network
 4. Transport
 5. Session
 6. Presentation
 7. Application

10. It is protocol dependent
11. It has 4 layer
12. The layers are
 1. Link
 2. Internet
 3. Transport
 4. Application

Example Network:

Internet:

The Internet has revolutionized many aspects of our daily lives. It has affected the way we do business as well as the way we spend our leisure time. Count the ways you've used the Internet recently. Perhaps you've sent electronic mail (e-mail) to a business associate, paid a utility bill, read a newspaper from a distant city, or looked up a local movie schedule-all by using the Internet.

Brief History

A network is a group of connected communicating devices such as computers and printers. An internet (note the lowercase letter i) is two or more networks that can communicate with each other. The most notable internet is called the Internet (uppercase letter I), a collaboration of more than hundreds of thousands of interconnected networks.

Private individuals as well as various organizations such as government agencies, schools, research facilities, corporations, and libraries in more than 100 countries use the Internet. Millions of people are users. Yet this extraordinary communication system only came into being in 1969.

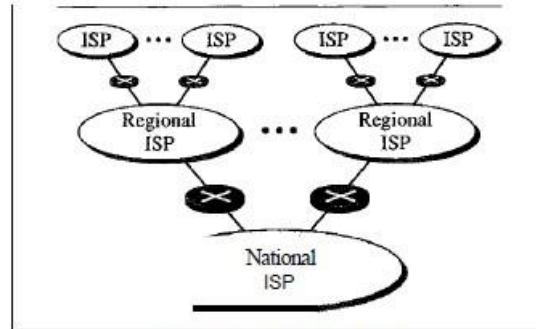
In the mid-1960s, mainframe computers in research organizations were standalone devices. Computers from different manufacturers were unable to communicate with one another. The Advanced Research Projects Agency (ARPA) in the Department of Defense (DoD) was interested in finding a way to connect computers so that the researchers they funded could share their findings, thereby reducing costs and eliminating duplication of effort.

In 1967, at an Association for Computing Machinery (ACM) meeting, ARPA presented its ideas for ARPANET, a small network of connected computers. The idea was that each host computer (not necessarily from the same manufacturer) would be attached to a specialized computer, called an **Interface Message Processor (IMP)**. The IMPs, in turn, would be connected to one another. Each IMP had to be able to communicate with other IMPs as well as with its own attached host. By 1969, ARPANET was a reality. Four nodes, at the University of California at Los Angeles (UCLA), the University of California at Santa Barbara (UCSB), Stanford Research Institute (SRI), and the University of Utah, were connected via the IMPs to form a network. Software called the **Network Control Protocol (NCP)** provided communication between the hosts.

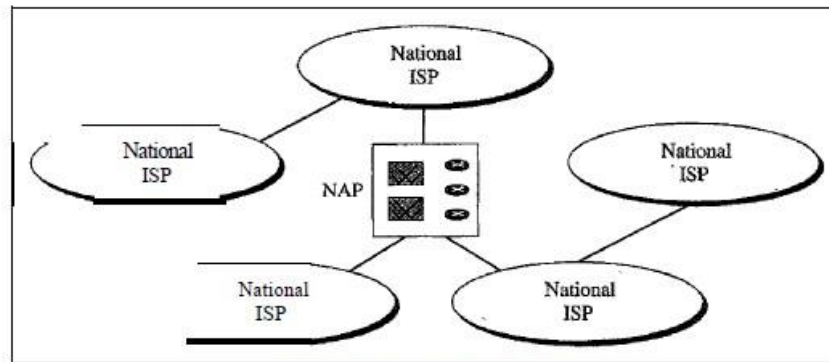
In 1972, Vint Cerf and Bob Kahn, both of whom were part of the core ARPANET group, collaborated on what they called the *Internetting Project*. Cerf and Kahn's landmark 1973 paper outlined the protocols to achieve end-to-end delivery of packets. This paper on Transmission Control Protocol (TCP) included concepts such as encapsulation, the datagram, and the functions of a gateway. Shortly thereafter, authorities made a decision to split TCP into two protocols: Transmission Control Protocol (TCP) and Internetworking Protocol (IP). IP would handle datagram routing while TCP would be responsible for higher-level functions such as segmentation, reassembly, and error detection. The internetworking protocol became known as TCP/IP.

The Internet has come a long way since the 1960s. The Internet today is not a simple hierarchical structure. It is made up of many wide- and local-area networks joined by connecting devices and switching stations. It is difficult to give an accurate representation of the Internet because it is continually changing-new networks are being added, existing networks are adding addresses, and networks of defunct companies are being removed.

Today most end users who want Internet connection use the services of Internet service providers (ISPs). There are international service providers, national service providers, regional service providers, and local service providers. The Internet today is run by private companies, not the government.



a. Structure of a national ISP



b. Interconnection of national ISPs

International Internet Service Providers

At the top of the hierarchy are the international service providers that connect nations together.

National Internet Service Providers

The national Internet service providers are backbone networks created and maintained by specialized companies. There are many national ISPs operating in North America; some of the most well-known are SprintLink, PSINet, UUNet Technology, AGIS, and internet Mel. To provide connectivity between the end users, these backbone networks are connected by complex switching stations (normally run by a third party) called network access points (NAPs). Some national ISP networks are also connected to one another by private switching stations called *peering points*. These normally operate at a high data rate (up to 600 Mbps).

Regional Internet Service Providers

Regional internet service providers or regional ISPs are smaller ISPs that are connected to one or more national ISPs. They are at the third level of the hierarchy with a smaller data rate.

Local Internet Service Providers

Local Internet service providers provide direct service to the end users. The local ISPs can be connected to regional ISPs or directly to national ISPs. Most end users are connected to the local ISPs. Note that in this sense, a local ISP can be a company that just provides Internet services, a corporation with a network that supplies services to its own employees, or a nonprofit organization,

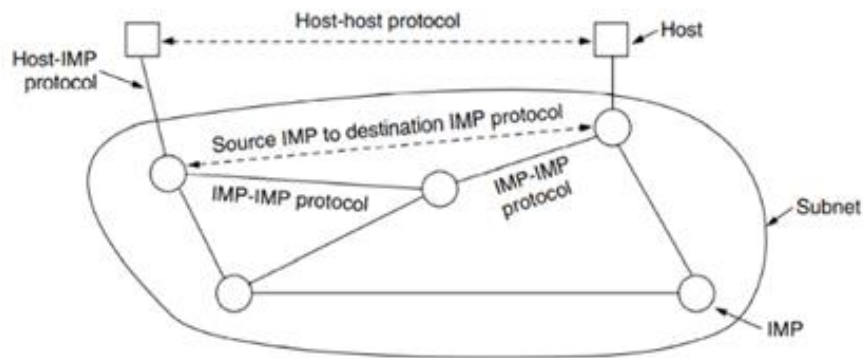
such as a college or a university, that runs its own network. Each of these local ISPs can be connected to a regional or national service provider.

Advanced Research Projects Agency Network (ARPANET)

- The Advanced Research Projects Agency Network (ARPANET) is a predecessor to the modern Internet. It was conceptualized in the 1950s, when computer scientists needed something better than the then available but unreliable switching nodes and network links.
- There were also only a limited number of large, powerful research computers, and researchers with access were separated geographically.
- The Advanced Research Projects Agency (ARPA) commissioned the development of an advanced and reliable way to connect these computers through a newly devised packet switching network, which was known as ARPANET.
- The ARPANET was a project funded by the U.S. government during the Cold War with USSR, in order to build a robust and reliable communications network. This was done by connecting various computers that could simultaneously communicate in a network that would not go down and continue running when a single node was taken out.
- The initial groundwork for a computer network was laid by BBN with other associates in 1963. And they were able to connect three computer terminals ARPA-sponsored computers:
 - i. The System Development Corporation (SDC) Q-32 at Santa Monica
 - ii. Project Genie at the University of California, Berkeley
 - iii. Multics at the Massachusetts Institute of Technology
- When Taylor needed to talk to someone at another computer, he would transfer to a different terminal for each connection. This was frustrating and led to the concept of one terminal/computer connected to a number of other terminals. This idea paved the way for the ARPANET and, eventually, the modern Internet.
- Paul Baran of Rand Corporation concluded that the strongest kind of network would be a packet switched network that would use any available communication line, regardless of the status of other lines. The ARPANET originally connected four computers, as follows:

- i. A Honeywell DDP 516 computer at University of California, Los Angeles
 - ii. An SDS-940 computer at the Stanford Research Institute
 - iii. An IBM 360/75 at University of California, Santa Barbara
 - iv. A DEC PDP-10 at the University of Utah
- Compatibility issues surfaces as more computers were connected to the network. These problems were solved in 1982 through the development of Transfer Control Protocol/Internet Protocol (TCP/IP).

Early ARPANET Architecture

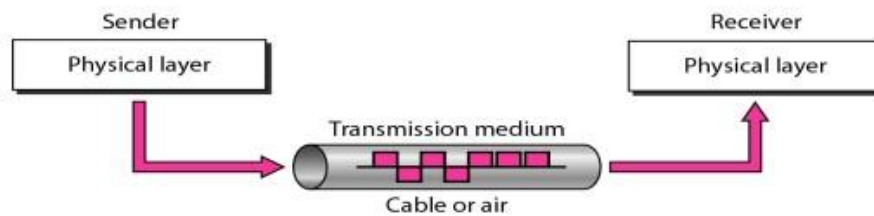


- BBN, a consulting firm build the subnet and wrote the subnet software. The software was split into two parts: subnet and host.
- The subnet software consisted of the IMP end of the host-IMP connection, the IMP-IMP protocol, and a source IMP to destination IMP protocol designed to improve reliability.
- Outside the subnet, software was also needed, namely, the host end of the host-IMP connection, the host-host protocol, and the application software.

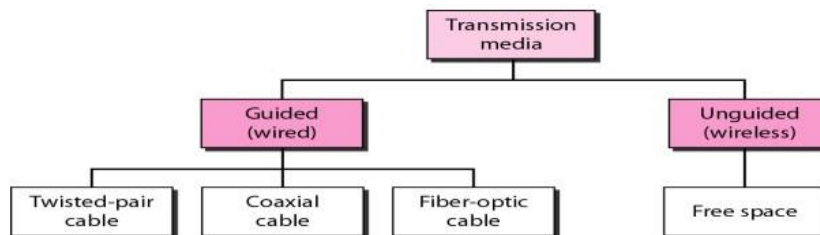
Physical Layer

Transmission media:

The main objective of the physical layer is to move data in the form of electromagnetic signals across a transmission medium.



The transmission medium is the physical path between transmitter and receiver in a data transmission system.



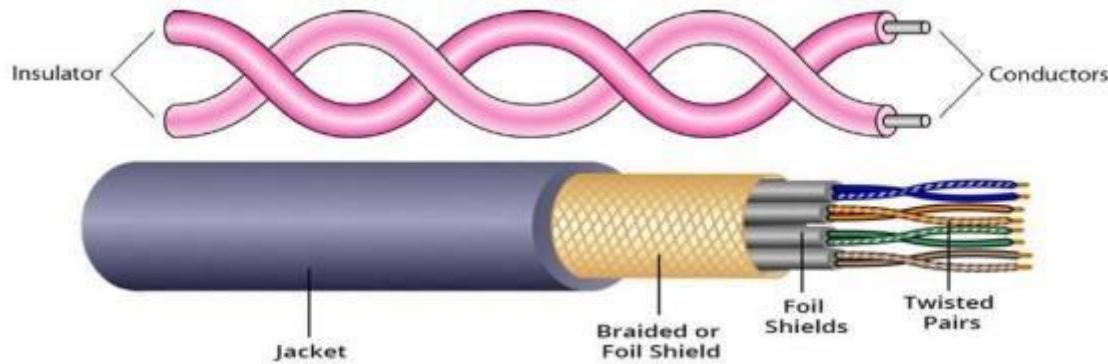
The signal travelling along any of these media is directed and contained by the physical limits of the medium.

Guided transmission medium can be used by the following:

1. Twisted pair cable
2. Co – axial cable
3. Optical fiber cable

1. Twisted Pair Cable:

Two insulated wires are twisted around each other and combined with others into a cable.



One of the wire is used to carry signals to the receiver and the other is used to only as a ground reference. In addition signal sent by the sender on one of the wires, interference (noise) and cross talk may affect both wires and created unwanted signals.

If the two wires are parallel, the effect of these unwanted signals is not the same in both wires because they are at different locations relative to the noised or cross talk sources, which may result in a difference at the receiver.

By using twisted pair, a balanced is maintained. These are used to connect telephone subscribers two switching centres and for wiring local area networks.

Twisted pair cables are two types:

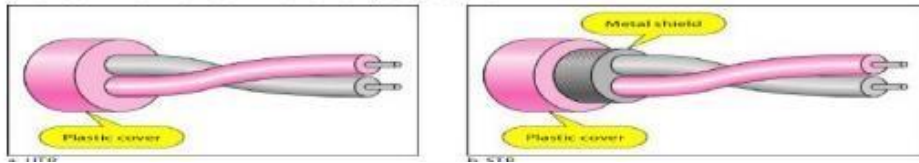
1. Unshielded TP (UTP)
2. Shielded TP (STP)

Most twisted pair cables used in communication is referred to as unshielded twisted pair (UTP). STP cable has a metal foil or braided mesh covering that encases each pair of insulated conductors. The metal casing improves the quality of cable by preventing the penetration of noise or cross talk.

STPs are bulkier more expensive

The most common UTP connector is RJ-45

It is a key connector inserted in only one way.



Advantages of twisted pair cable

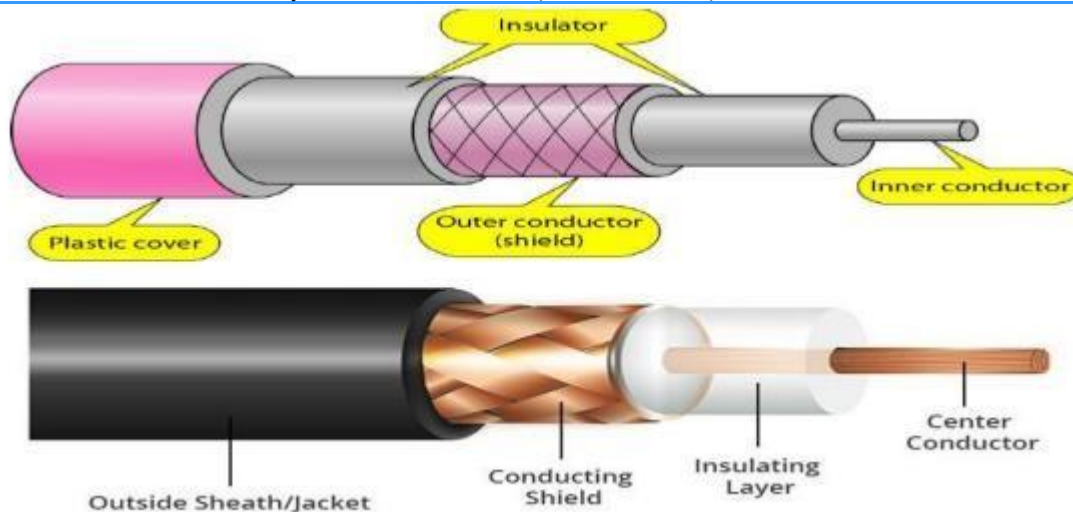
- Connections can be easy and very flexible.
- Trained personnel easily available easy to install and low maintenance
- Adequate performance
- Can be used for both analog and digital transmissions
- Less expensive for short distances

Disadvantages of twisted pair cable

- Due to attenuation(losing signal strength over distance) Signal cannot travel long distances without repeaters
- Low Bandwidth compared to other Media, Usual data transmission rate ranges from 1 Mbps to 10 Mbps.
- High error rate for distances greater than 100m
- Very thin and hence breaks easily
- Not suitable for broadband connections

2. Co – Axial Cable:

Co – axial cable carries signals of higher frequency ranges than those in twisted pair cable. It is like a twisted pair, it has two conductors that are shielded.



The outer metallic wrapping serves both as a shield against noise and is the second conductor, which completes the circuit. The outer conductor is also enclosed in an insulating sheath and whole cable is protected by plastic cover.

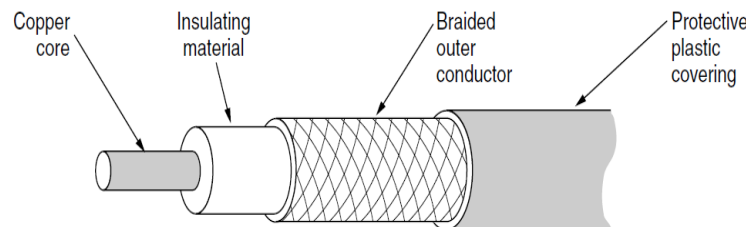
Co – axial cable uses for TV supports a spectrum of 50 – 750 MHZ.

RG – 59 – 75 Ω of impedance – cable TV.

RG – 58 – 50 Ω – thin Ethernet.

RG – 11 – 50 Ω - thick Ethernet

Co – axial cable could carry digital data up to 600 mbps.



Types of Coaxial cables

There are Two Types of Coaxial Cables

- **Baseband Coaxial Cable:** LAN generally uses Baseband Coaxial cable. It is the 50 ohm coaxial cable used for digital transmission. The cable comes with a power of transmitting a single signal at quite high speed. It transmits a single signal at a time.
- **Broadband Coaxial Cable:** This cable transmits many simultaneous signals and that too using different frequency. It covers more area than the Baseband coaxial cable and can

run nearly 100km, but requires analog amplifiers. Amplifiers are used to strengthen the signal periodically.

Advantages and Disadvantages of Coaxial Cables

Advantages of Coaxial Cables:

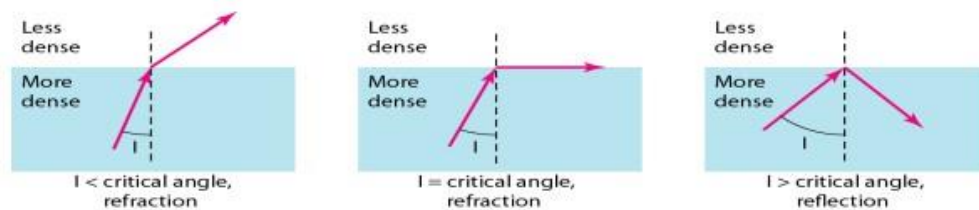
- Excellent noise immunity
- Signals can travel longer distances at higher speeds, e.g. 1 to 2 Gbps for 1 Km cable
- Can be used for both analog and digital signals
- Inexpensive as compared to fiber optic cables
- Easy to install and maintain

Disadvantages of Coaxial Cables:

- Expensive as compared to twisted pair cables
- Not compatible with twisted pair cables

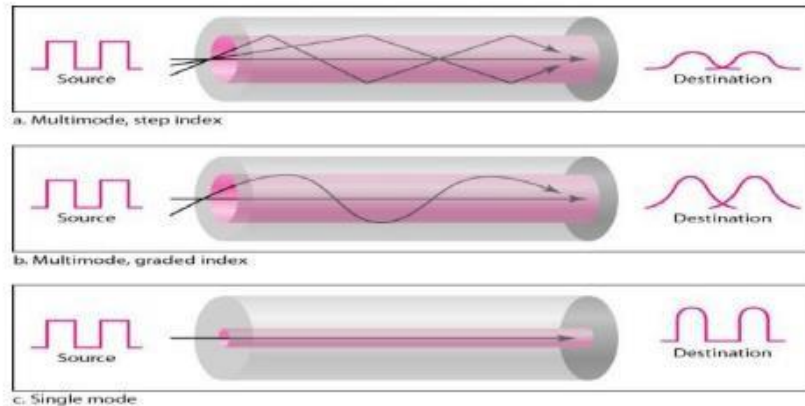
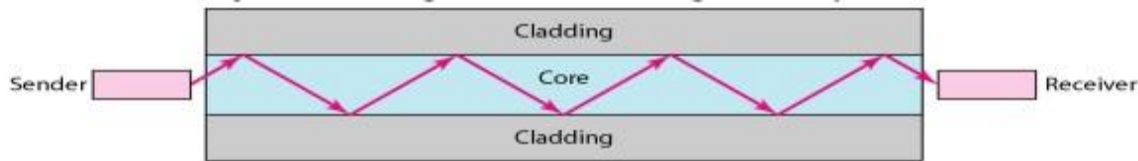
3. Fiber Optic Cable:

A fiber optic cable is made of glass or plastic and transmits signals in the form of light. The light travels in a straight line as long as it is moving through a single uniform substance. If a ray of light travelling through one substance suddenly enters another substance, the ray changes direction.

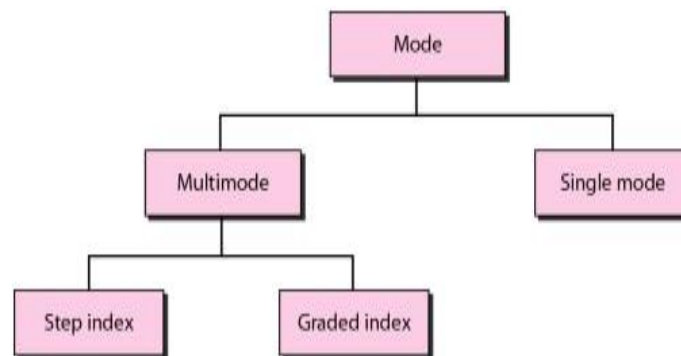


Above figure shows if the angle of incidence I is $<$ the critical angle, the ray refracts and moves closer to the surface. If angle of incidence I is equal the critical angle, the light bends along the interface. If angle of incidence I is $>$ the critical angle, the ray reflects and travels again in the denser substance.

Optical fibers use reflection to guide light through a channel. A glass or plastic core is surrounded by a cladding of less denser glass or plastic. The difference in density of the two materials must be such that a beam of light moving through the core is reflected off the cladding instead of being refracted into it.



Modes of Propagation:

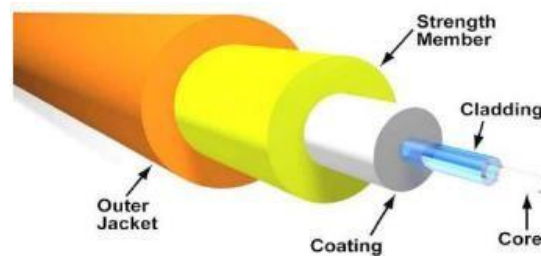


In multimode, the multiple beams from a light source move through the core in different patterns. In multi-mode step index, the density of the core remains constant from the centre of the edges. The beam of light moves through this constant density in a straight line until it reaches the interface of the core and the cladding. At the interface, there is abrupt changes due to a lower density.

In multimode step index fiber, the suddenness of this change, which contributes to the distortion of the signal as it passes through the fiber.

In multimode graded index fiber, decreases the distortion of the signal through the cable. A graded index fiber is one with varying densities. Density is highest at the center of the core and decreases gradually to its lowest at the edge.

In single mode, we may use step index fiber and a highly focuses sources of light that limits beam to a small range of angles, all close to the horizontal. It is made with smaller diameter than that of the multimode fiber.



Advantages & Disadvantages:

Advantages:

- Increased capacity and bandwidth
- Light weight
- Less signal attenuation
- Immunity to electromagnetic interference
- Resistance to corrosive materials

Disadvantages:

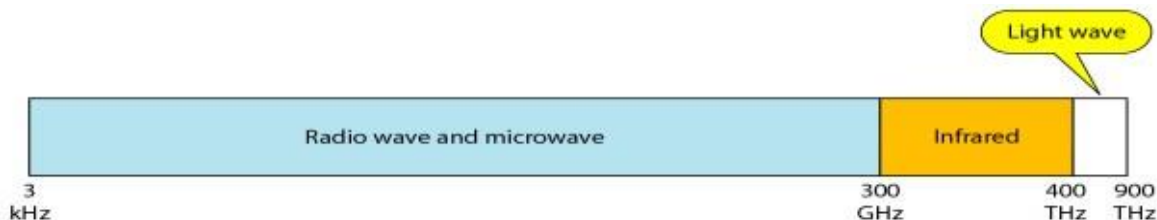
Difficult to install and maintain

- High cost
- Fragile

Un-Guided Transmission(Wireless Transmission Media):

Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is called as wireless communications

Signals are normally broad cast through free space and thus are available to anyone. Who has a device capable of receiving them.



Unguided signals can travel from the source to destination in several ways.

- i) Groundwave propagation
- ii) Sky wave propagation
- iii) Line – of – sight propagation

i) Ground Wave Propagation

In ground wave propagation, radio wave travels through the lowest portion of the atmosphere, touches the earth. These low frequency signals propagate in all directions from the transmitting antenna and follow the curvature of the planet.

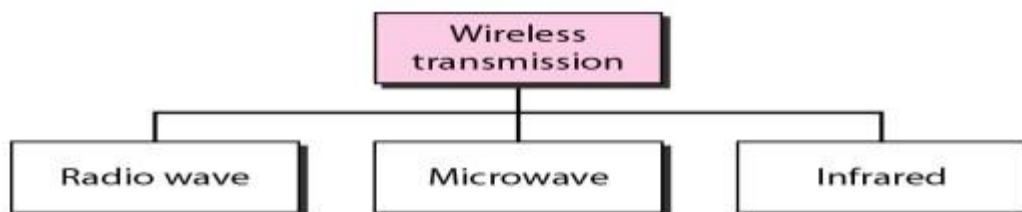
ii) Sky Wave Propagation

In sky wave propagation, higher frequency radio waves radiate upwards into the ionosphere where they are reflected back to earth. This type of transmission allows for greater distances with lower output power.

iii) Line – Sight Propagation

In line – of – sight propagation, very high frequency signals are transmitted in straight lines directly from antenna to antenna.

Antenna must be directional, facing each other and either tall enough or close enough together not to be affected by the curvature of earth.



1. Radio Waves:

In an electromagnetic spectrum, electron magnetic waves ranging in frequencies between 3 KHZ and 1 GHZ are normally called radio waves.

Radio waves are Omni directional, when an antenna transmits radio waves, they are propagated in all directions.

Radio wave particularly propagates in the sky mode and can travel long distances and it is good for broad casting.

Radio waves use Omni directional antennas that send out signals in all directions. Radio waves are used for multicast communications, such as radio and television, and paging systems.

2. Micro Waves:

Electromagnetic waves having frequencies between 1 and 300 GHZ are called micro waves

Micro waves are Uni-directional, when an antenna transmits micro waves, they can be narrowly focused.

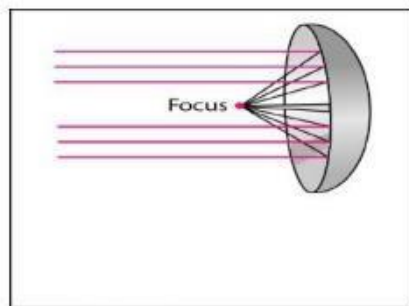
Micro wave propagation is line – of – sight.

Repeaters are often needed for long distance communication.

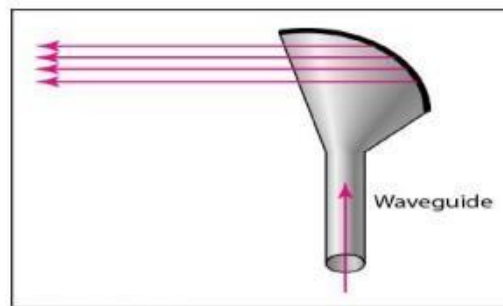
Very light frequency micro waves cannot penetrate walls.

Micro waves need Uni-directional antennas that send out signals in one direction.

Two types antennas used in microwave communication parabolic dish antenna horn antenna.



a. Dish antenna



b. Horn antenna

Microwaves are used for unicast communication such as cellular telephones, satellite networks, and wireless LANs.

3. Infrared Waves:

Infrared waves with frequencies from 300 GHZ to 400 GHZ (wave length from 1 mm to 770 mm) can be used for short – range communications.

Infrared waves, having high frequencies, cannot penetrate.

Infrared signals can be used for short-range communication in a closed area using line-of sight propagation.

Switching:

Switching is process to forward packets coming in from one port to a port leading towards the destination. When data comes on a port it is called ingress, and when data leaves a port or goes out it is called egress. A communication system may include number of switches and nodes. At broad level, switching can be divided into two major categories:

- **Connectionless (Packet Switching):** The data is forwarded on behalf of forwarding tables. No previous handshaking is required and acknowledgements are optional.
- **Connection Oriented (Circuit Switching):** Before switching data to be forwarded to destination, there is a need to pre-establish circuit along the path between both endpoints. Data is then forwarded on that circuit. After the transfer is completed, circuits can be kept for future use or can be turned down immediately.

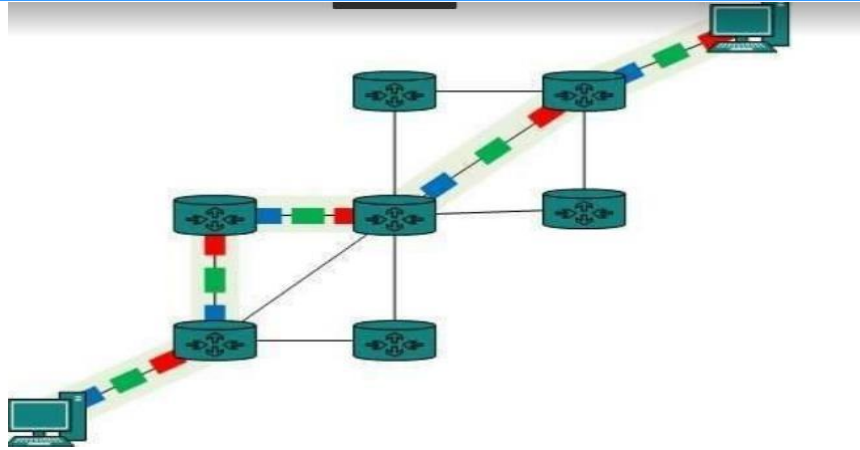
Circuit Switching:

When two nodes communicate with each other over a dedicated communication path, it is called circuit switching. There is a need of pre-specified route from which data will travel and no other data is permitted. In circuit switching, to transfer the data, circuit must be established so that the data transfer can take place.

Circuits can be permanent or temporary. Applications which use circuit switching may have to go through three phases:

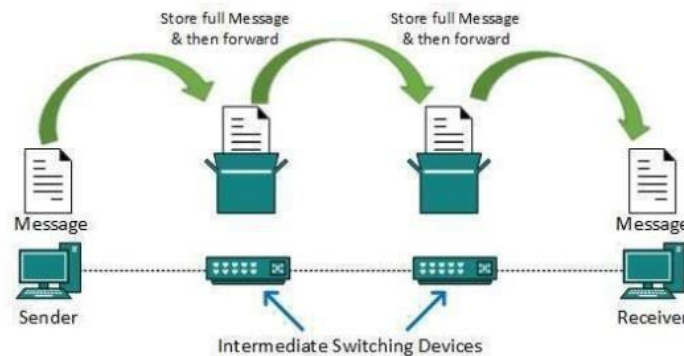
- Establish a circuit
- Transfer the data
- Disconnect the circuit

Circuit switching was designed for voice applications. Telephone is the best suitable example of circuit switching. Before a user can make a call, a virtual path between caller and callee is established over the network.



Message Switching

This technique was somewhere in middle of circuit switching and packet switching. In message switching, the whole message is treated as a data unit and is switching / transferred in its entirety. A switch working on message switching, first receives the whole message and buffers it until there are resources available to transfer it to the next hop. If the next hop is not having enough resource to accommodate large size message, the message is stored and switch waits.



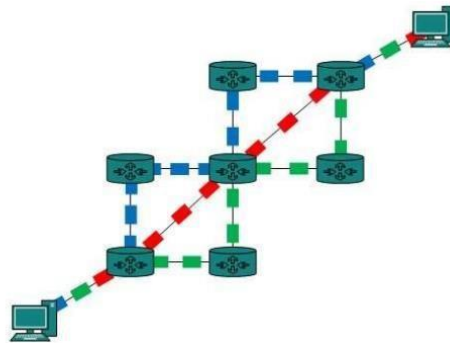
This technique was considered substitute to circuit switching. As in circuit switching the whole path is blocked for two entities only. Message switching is replaced by packet switching. Message switching has the following drawbacks:

- Every switch in transit path needs enough storage to accommodate entire message.
- Because of store-and-forward technique and waits included until resources are available, message switching is very slow.
- Message switching was not a solution for streaming media and real-time applications.

Packet Switching

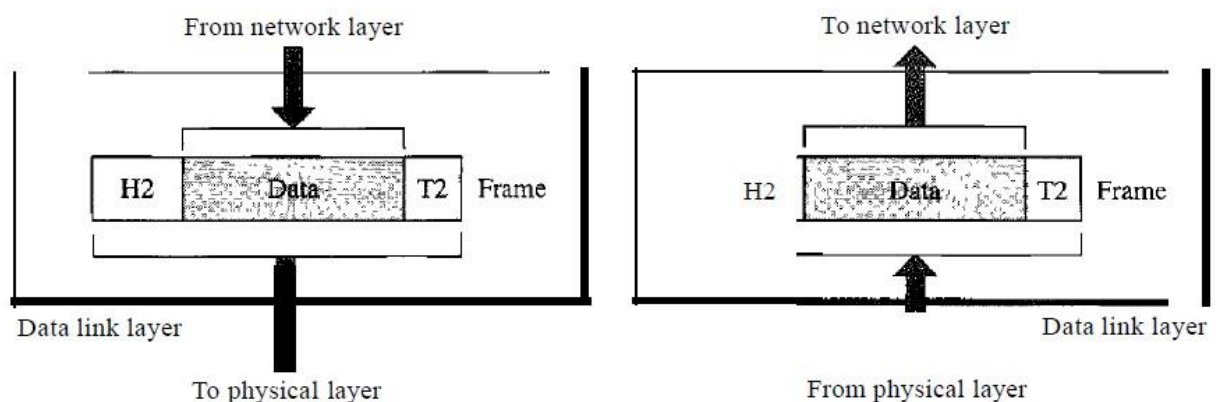
Shortcomings of message switching gave birth to an idea of packet switching. The entire message is broken down into smaller chunks called packets. The switching information is added in the header of each packet and transmitted independently. It is easier for intermediate networking devices to store small size packets and they do not take much resources either on carrier path or in the internal memory of switches.

Packet switching enhances line efficiency as packets from multiple applications can be multiplexed over the carrier. The internet uses packet switching technique. Packet switching enables the user to differentiate data streams based on priorities. Packets are stored and forwarded according to their priority to provide quality of service.



Data Link Layer – Design Issues:

The data link layer transforms the physical layer, and raw information's is transmitted to a reliable link. It make physical layer appear error – free to the upper layer i.e. network layer.



Above figure shows that relationship between the data link layer to n/w layer and physical layer.

The responsibility of the data link layer is to move frames one hop (node) to the next hop.

The data link layer uses the services of the physical layer to send and receive bits over the communication channel.

1. It has to provide well – defined service interface to network layer.
2. Dealing with transmission errors.
3. Regulating the flow of data so that slow receiver are not swamped by the fast senders.

Framing

Physical Addressing

If frames are to be distributed to different systems on the network, the data link layer adds a header to the frame to define the sender and or receiver of the frame.

Flow control

When a sender systematically wants to send frames faster than the receiver can accept them. The rate at which the data are absorbed by the receiver less than the rate at which the data are produced in the sender

The data link layer imposes a flow control mechanism to avoid over whelming the receiver.

Error Control

The data link layer adds reliability to the physical layer by adding mechanism to detect and retransmit damaged or lost frames.

- It also uses a mechanism to recognize duplicate frames
- Error control is normally achieved through a trailer added to the end of the frame.

Access Control

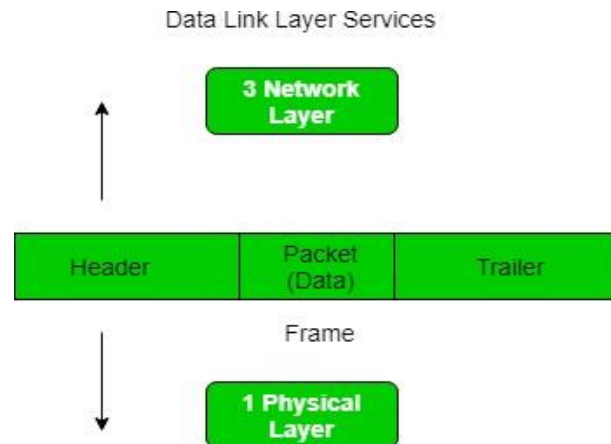
When two or more devices are connected to the same link, the data link layer protocols are necessary to determine which device has control over the link at any given time.

Design Issues

1. **Services to Network layer**
2. **Framing in Data Link Layer**

Frames are the units of digital transmission particularly in computer networks and telecommunications. Frames are comparable to the packets of energy called photons in case of light energy. Frame is continuously used in Time Division Multiplexing process. Framing is a

point-to-point connection between two computers or devices consists of a wire in which data is transmitted as a stream of bits. However, these bits must be framed into discernible blocks of information. Framing is a function of the data link layer. It provides a way for a sender to transmit a set of bits that are meaningful to the receiver. Ethernet, token ring, frame relay, and other data link layer technologies have their own frame structures. Frames have headers that contain information such as error-checking codes.



At data link layer, it extracts message from sender and provide it to receiver by providing sender's and receiver's address. The advantage of using frames is that data is broken up into recoverable chunks that can easily be checked for corruption.

Problems in Framing

- **Detecting start of the frame:** When a frame is transmitted, every station must be able to detect it. Station detect frames by looking out for special sequence of bits that marks the beginning of the frame i.e. SFD (Starting Frame Delimiter).
- **How do station detect a frame:** Every station listen to link for SFD pattern through a sequential circuit. If SFD is detected, sequential circuit alerts station. Station checks destination address to accept or reject frame.
- **Detecting end of frame:** When to stop reading the frame.

Types of framing – There are two types of framing:

1. Fixed size – The frame is of fixed size and there is no need to provide boundaries to the frame, length of the frame itself acts as delimiter.

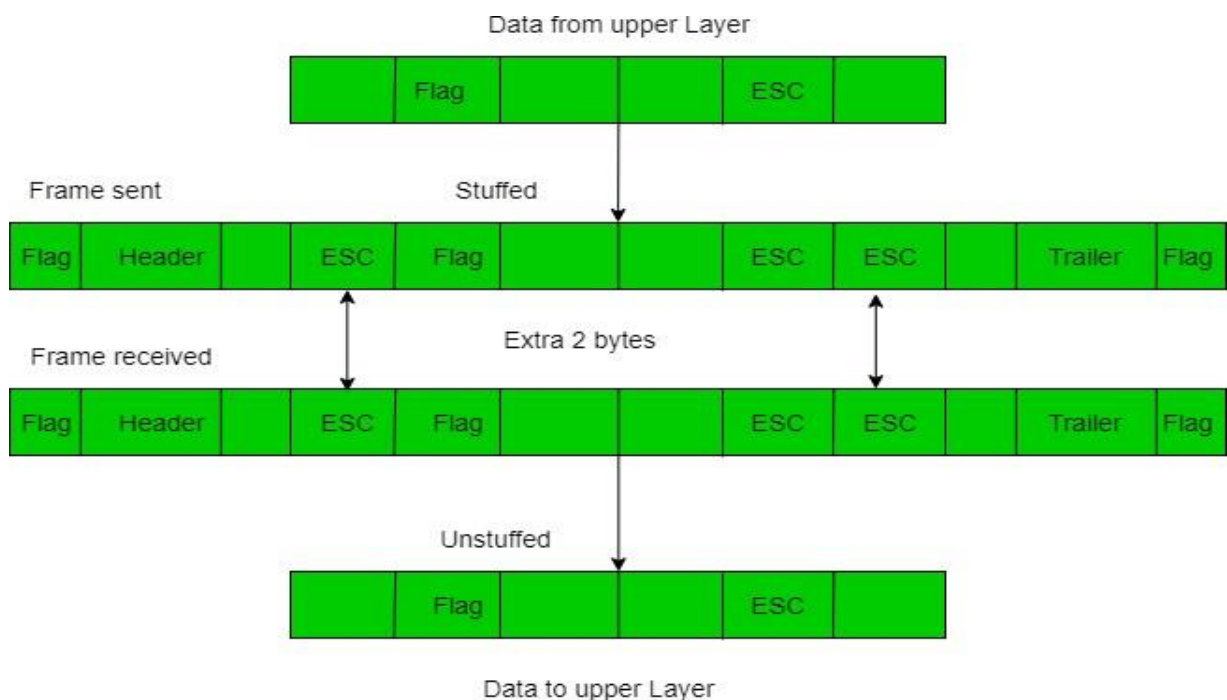
Drawback: It suffers from internal fragmentation if data size is less than frame size

Solution: Padding

3. Variable size – In this there is need to define end of frame as well as beginning of next frame to distinguish. This can be done in two ways:

- i. **Length field (Character Count)** – We can introduce a length field in the frame to indicate the length of the frame. Used in **Ethernet(802.3)**. The problem with this is that sometimes the length field might get corrupted.
- ii. **End Delimiter (ED)** – We can introduce an ED (pattern) to indicate the end of the frame. The problem with this is that ED can occur in the data. This can be solved by:
- iii. **Character/Byte Stuffing:** Used when frames consist of character. If data contains ED then, byte is stuffed into data to differentiate it from ED. Let ED = “\$” → if data contains ‘\$’ anywhere, it can be escaped using ‘\O’ character.

→ if data contains ‘\O\$’ then, use ‘\O\O\O\$’(\$ is escaped using \O and \O is escaped using \O).



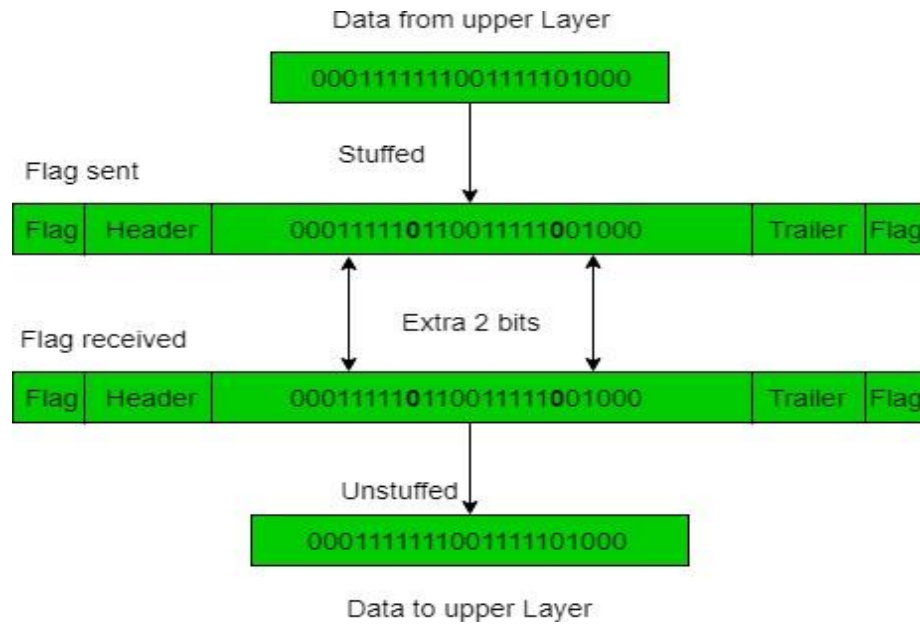
Disadvantage – It is very costly and obsolete method.

1. Bit Stuffing: Let ED = 01111 and if data = 01111

→ Sender stuffs a bit to break the pattern i.e. here appends a 0 in data = 0111**0**1.

→ Receiver receives the frame.

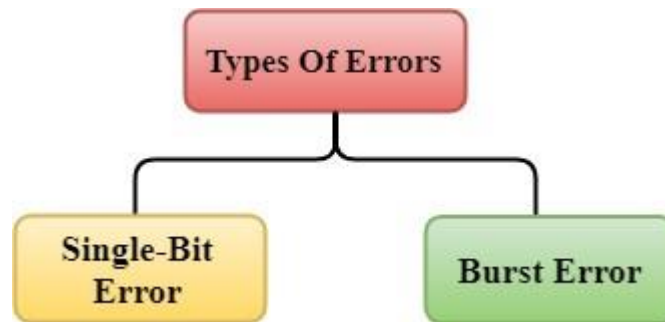
→ If data contains 011101, receiver removes the 0 and reads the data.



2. Error Detection & Correction

When data is transmitted from one device to another device, the system does not guarantee whether the data received by the device is identical to the data transmitted by another device. An Error is a situation when the message received at the receiver end is not identical to the message transmitted.

Types of Errors:



Errors can be classified into two categories:

- Single-Bit Error
- Burst Error

Single-Bit Error:

The only one bit of a given data unit is changed from 1 to 0 or from 0 to 1.

Burst Error:

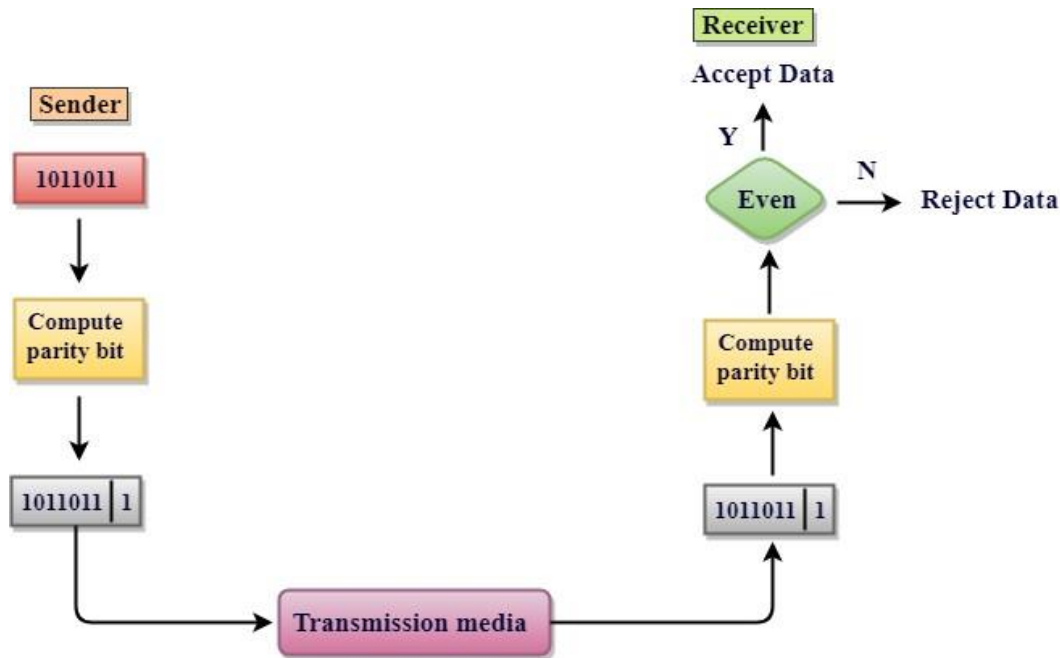
The two or more bits are changed from 0 to 1 or from 1 to 0 is known as Burst Error.

Error Detection Techniques:

i) Parity check

- Parity checking is the simple mechanism and inexpensive to detect the errors.
- In this technique, a redundant bit is also known as a parity bit which is appended at the end of the data unit so that the number of 1s becomes even. Therefore, the total number of transmitted bits would be 9 bits.
- If the number of 1s bits is odd, then parity bit 1 is appended and if the number of 1s bits is even, then parity bit 0 is appended at the end of the data unit.

- At the receiving end, the parity bit is calculated from the received data bits and compared with the received parity bit.
- This technique generates the total number of 1s even, so it is known as even-parity checking.



Drawbacks of Single Parity Checking

- It can only detect single-bit errors which are very rare.
- If two bits are interchanged, then it cannot detect the errors.

ii) Checksum:

This is a block code method where a checksum is created based on the data values in the data blocks to be transmitted using some algorithm and appended to the data. When the receiver gets this data, a new checksum is calculated and compared with the existing checksum. A non-match indicates an error.

Error Detection by Checksum

For error detection by checksum, data is divided into fixed sized frames or segments.

- **Sender's End** – The sender adds the segments using 1's complement arithmetic to get the sum. It then complements the sum to get the checksum and sends it along with the data frames.
- **Receiver's End** – The receiver adds the incoming segments along with the checksum using 1's complement arithmetic to get the sum and then complements it.
If the result is zero, the received frames are accepted; otherwise they are discarded.

Suppose that the sender wants to send 4 frames each of 8 bits, where the frames are 11001100, 10101010, 11110000 and 11000011.

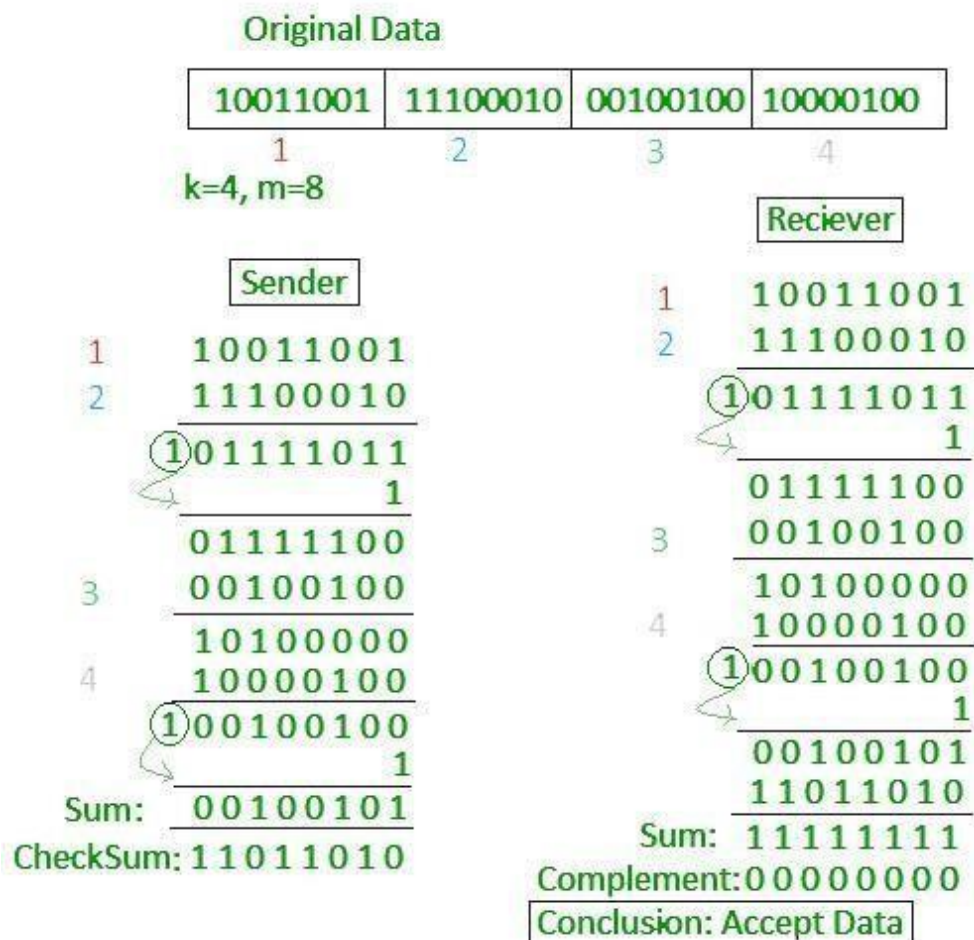
The sender adds the bits using 1s complement arithmetic. While adding two numbers using 1s complement arithmetic, if there is a carry over, it is added to the sum.

After adding all the 4 frames, the sender complements the sum to get the checksum, 11010011, and sends it along with the data frames.

The receiver performs 1s complement arithmetic sum of all the frames including the checksum. The result is complemented and found to be 0. Hence, the receiver assumes that no error has occurred.

Sender's End		Receiver's End	
Frame 1:	11001100	Frame 1:	11001100
Frame 2:	+ 10101010	Frame 2:	+ 10101010
Partial Sum:	1 01110110	Partial Sum:	1 01110110
	+ 1		+ 1
	01110111		01110111
Frame 3:	+ 11110000	Frame 3:	+ 11110000
Partial Sum:	1 01100111	Partial Sum:	1 01100111
	+ 1		+ 1
	01101000		01101000
Frame 4:	+ 11000011	Frame 4:	+ 11000011
Partial Sum:	1 00101011	Partial Sum:	1 00101011
	+ 1		+ 1
Sum:	00101100	Sum:	00101100
Checksum:	11010011	Checksum:	11010011
		Sum:	11111111
		Complement:	00000000
		Hence accept frames.	

- In checksum error detection scheme, the data is divided into k segments each of m bits.
- In the sender's end the segments are added using 1's complement arithmetic to get the sum. The sum is complemented to get the checksum.
- The checksum segment is sent along with the data segments.
- At the receiver's end, all received segments are added using 1's complement arithmetic to get the sum. The sum is complemented.
- If the result is zero, the received data is accepted; otherwise discarded.



iii) Cyclic Redundancy Check Codes:

In order to send the data from sender to receiver, the data must be checked and preceded with zero errors to receiver. CRC (cyclic redundancy check) is also called as polynomial code.

Polynomial codes are based on the treating bit string as representations of polynomials with coefficients of 0 and 1 only.

It is used in networks such as LANs and WANs.

In sender or encoder, the generator has k bits (4 – bits) the code word has n bits (7 – bits) are given by as per the structure.

Generator is : 1 1 0 1

Data word is : 1 0 1 1 0 1 1

The size of the data word is appended by $n - k$ (bits) 0s to the right hand side of the code word.

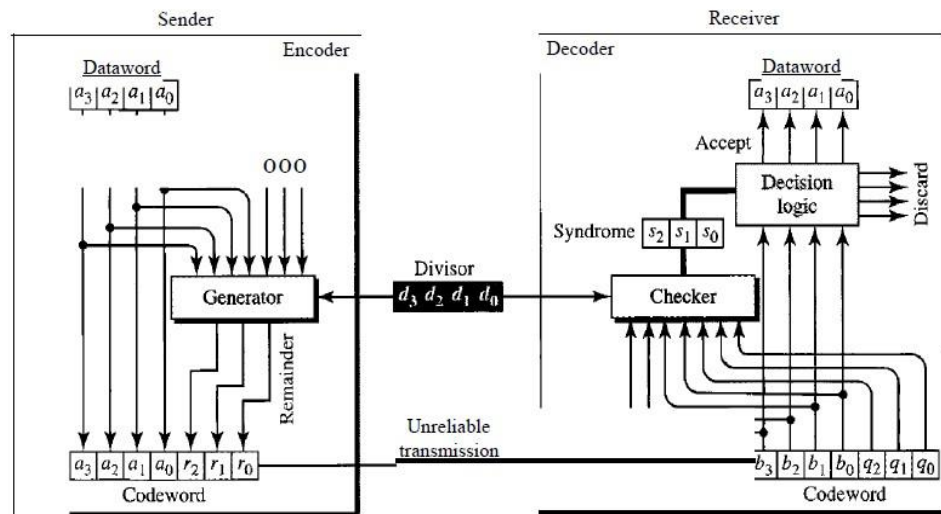
The CRC generator has 4 – bits, so (4-1) three zeros has to be added to data word. Therefore, the code word is given by: 1 0 1 1 0 1 1 0 0 0

The generator divides the augmented data word by divisor (modulo – 2 divisions), the quotient of the division is discarded.

The remainder is appended to the data word to create the code word.

A copy of all $n - k$ bits is fed to the checker which is replica of the generator.

The remainder produced by the checker is a syndrome of $n - k$ (3 – bits) which is fed to decision logic analyzer.



The analyzer has to perform simple function like if the syndrome bits are all 0s, the 4 left most bits of the code word are accepted as the data word (interpreted as no error) otherwise the bits are non – zero mean (interpreted as error) and are discarded.

Example:

Perform CRC check for error free data transmission to receiver. generator is: 1 1 0 1 and data word is given by: 1 0 1 1 0 1 1.

Solution:

Generator : 1 1 0 1 (k – bits)

Dataword : 1 0 1 1 0 1 1 (n – bits)

Cyclic redundancy generators is: $(k-1) = 0 0 0$ (3 – bits)

The 3 – bits are append to the data word to check CRC. Perform a division operation with code word. i.e., by modulo – 2 division it also represents EX – OR like operation

$$\begin{array}{r}
 111 \\
 \hline
 1101 \overline{) 1011011000} \\
 1101 \\
 \hline
 0 10011000 \\
 1 101 \\
 \hline
 0 1111000 \\
 1 101 \\
 \hline
 0 1100 \\
 1 101 \\
 \hline
 0 001 \rightarrow \text{CRC - bit}
 \end{array}$$

Case 1:

In the above operation last bit is '1' the CRC both is append to the code word to check error free transmission.

Checking at the receiver side

$$\begin{array}{r}
 1 111 \\
 1101 \\
 \hline
 0 110011001 \\
 1 101 \\
 \hline
 0 1111001 \\
 1 101 \\
 \hline
 \hline
 1101 \overline{) 1011011001}
 \end{array}$$

$$\begin{array}{rcl}
 & 0 & \text{0 1 1 0 1} \\
 & 1 & \text{1 0 1} \\
 0 & 0 0 0 & \rightarrow \overline{\text{Syndrome is '0'}.}
 \end{array}$$

If code word have a corrupted bit, for such cases single error may occur, which will be discarded.

Case 2:

1 1 0 1) 1 0 **1 1** 1 0 1 1 0 0 1 (

0

Corrupted bit – replaced as ‘0’

1 1

1 1 0 1) 1 0 0 1 0 1 1 0 0 1 (

1 1 0 1

0 1 0 0 0 1 1 0 0 1

1 1 0 1

0 1 0 1 1 1 0 0 1

1 1 0 1

0 1 1 0 1 0 0 1

1 1 0 1

~~0 0 0 0 0 0 1~~ → Syndrome is ‘1’ □ Error Bit

In this case data word not accepted, discarded due to non – zero CRC – bit occur. For an successful transmission of data is selected by the checking of CRC must done with above manner.

Error Correction Code

Hamming code is a set of error-correction codes that can be used to detect and correct the errors that can occur when the data is moved or stored from the sender to the receiver. It is a technique developed by R.W. Hamming for error correction.

The number of redundant bits can be calculated using the following formula:

$$2^r \geq m + r + 1$$

where, r = redundant bit, m = data bit

General Algorithm of Hamming code: Hamming Code is simply the use of extra parity bits to allow the identification of an error.

1. Write the bit positions starting from 1 in binary form (1, 10, 11, 100, etc).
2. All the bit positions that are a power of 2 are marked as parity bits (1, 2, 4, 8, etc).
3. All the other bit positions are marked as data bits.
4. Each data bit is included in a unique set of parity bits, as determined its bit position in binary form.
 - a. Parity bit 1 covers all the bits positions whose binary representation includes a 1 in the least significant position (1, 3, 5, 7, 9, 11, etc).
 - b. Parity bit 2 covers all the bits positions whose binary representation includes a 1 in the second position from the least significant bit (2, 3, 6, 7, 10, 11, etc).
 - c. Parity bit 4 covers all the bits positions whose binary representation includes a 1 in the third position from the least significant bit (4–7, 12–15, 20–23, etc)
 - d. Parity bit 8 covers all the bits positions whose binary representation includes a 1 in the fourth position from the least significant bit bits (8–15, 24–31, 40–47, etc).
 - e. In general, each parity bit covers all bits where the bitwise AND of the parity position and the bit position is non-zero.
1. Since we check for even parity set a parity bit to 1 if the total number of ones in the positions it checks is odd.
2. Set a parity bit to 0 if the total number of ones in the positions it checks is even.

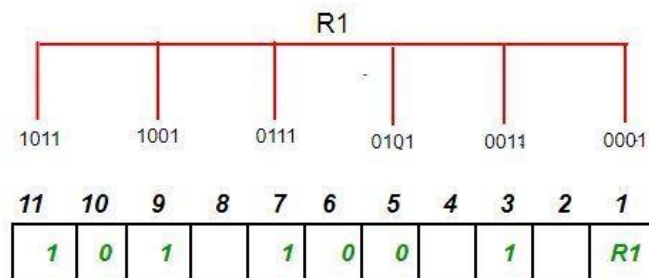
Determining the position of redundant bits – These redundancy bits are placed at positions that correspond to the power of 2.

Suppose the data to be transmitted is 1011001, the bits will be placed as follows:

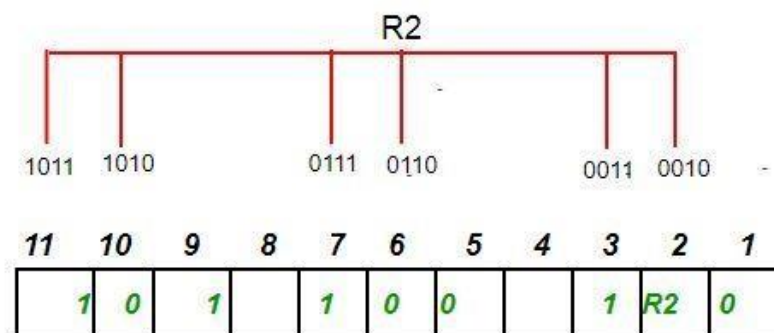
11	10	9	8	7	6	5	4	3	2	1
1	0	1	R8	1	0	0	R4	1	R2	R1

Determining the Parity bits:

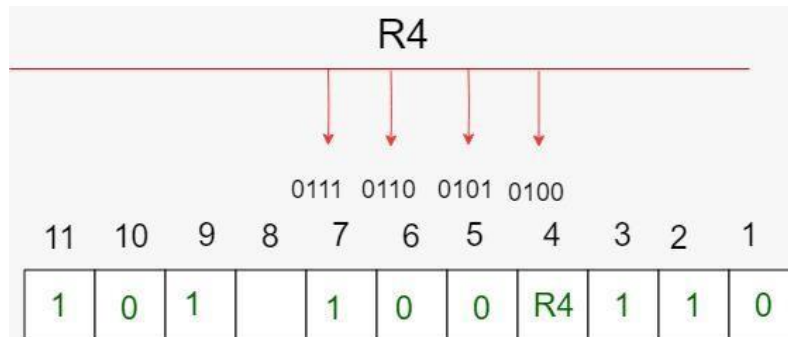
R1 bit is calculated using parity check at all the bits positions whose binary representation includes a 1 in the least significant position. R1: bits 1, 3, 5, 7, 9, 11



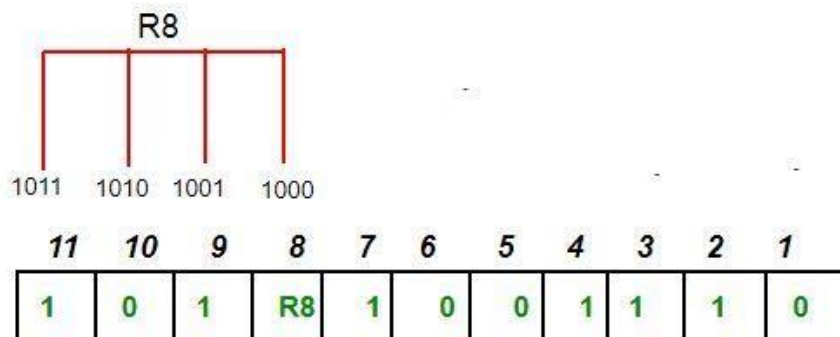
- To find the redundant bit R1, we check for even parity. Since the total number of 1's in all the bit positions corresponding to R1 is an even number the value of R1 (parity bit's value) = 0
- R2 bit is calculated using parity check at all the bits positions whose binary representation includes a 1 in the second position from the least significant bit. R2: bits 2,3,6,7,10,11



- To find the redundant bit R2, we check for even parity. Since the total number of 1's in all the bit positions corresponding to R2 is odd the value of R2 (parity bit's value) = 1
- R4 bit is calculated using parity check at all the bits positions whose binary representation includes a 1 in the third position from the least significant bit. R4: bits 4, 5, 6, 7



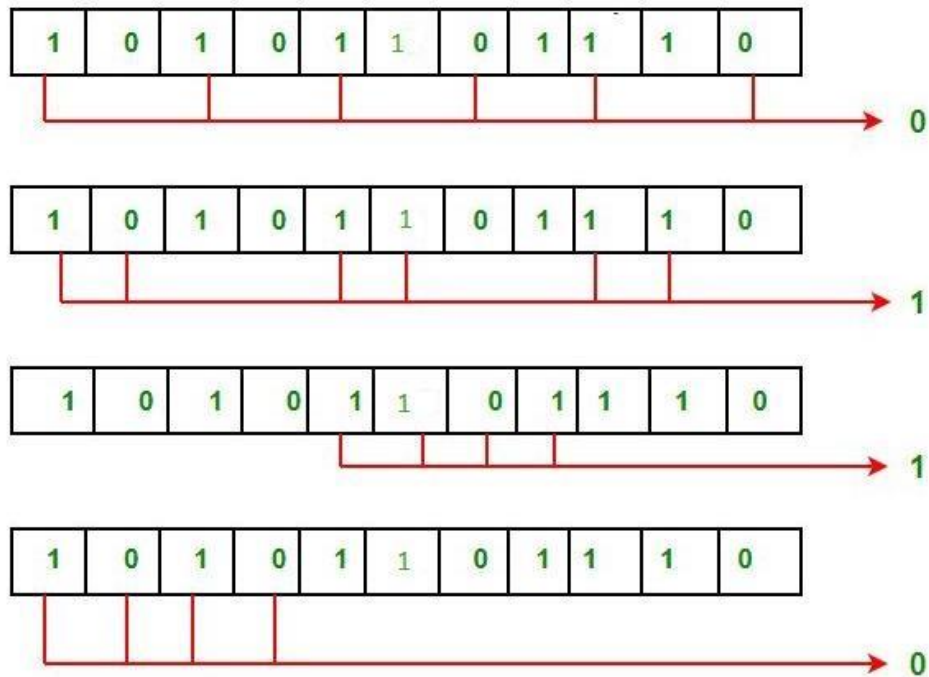
- To find the redundant bit R4, we check for even parity. Since the total number of 1's in all the bit positions corresponding to R4 is odd the value of R4 (parity bit's value) = 1
- R8 bit is calculated using parity check at all the bits positions whose binary representation includes a 1 in the fourth position from the least significant bit. R8: 8,9,10,11. bit



- To find the redundant bit R8, we check for even parity. Since the total number of 1's in all the bit positions corresponding to R8 is an even number the value of R8 (parity bit's value) = 0. Thus, the data transferred is:

11	10	9	8	7	6	5	4	3	2	1
1	0	1	0	1	0	0	1	1	1	0

Error detection and correction: Suppose in the above example the 6th bit is changed from 0 to 1 during data transmission, then it gives new parity values in the binary number:



The bits give the binary number 0110 whose decimal representation is 6. Thus, bit 6 contains an error. To correct the error the 6th bit is changed from 1 to 0.